

APP設計實習

1142數位教材

單元3:圖像處理

主講老師：徐培倫



單元3

圖像處理

- ◆ Android 模擬器安裝
- ◆ 雷電模擬器LDplayer
 - AI Companion
- ◆ 手機 APP 實例操作畫面
 - Hello 隱藏及顯示圖像
 - HelloTest 顯示另一張圖像
- ◆ 圖像切換功能
- ◆ 作業：展示Show





Android 模擬器安裝

◆ 3種方法測試Android App

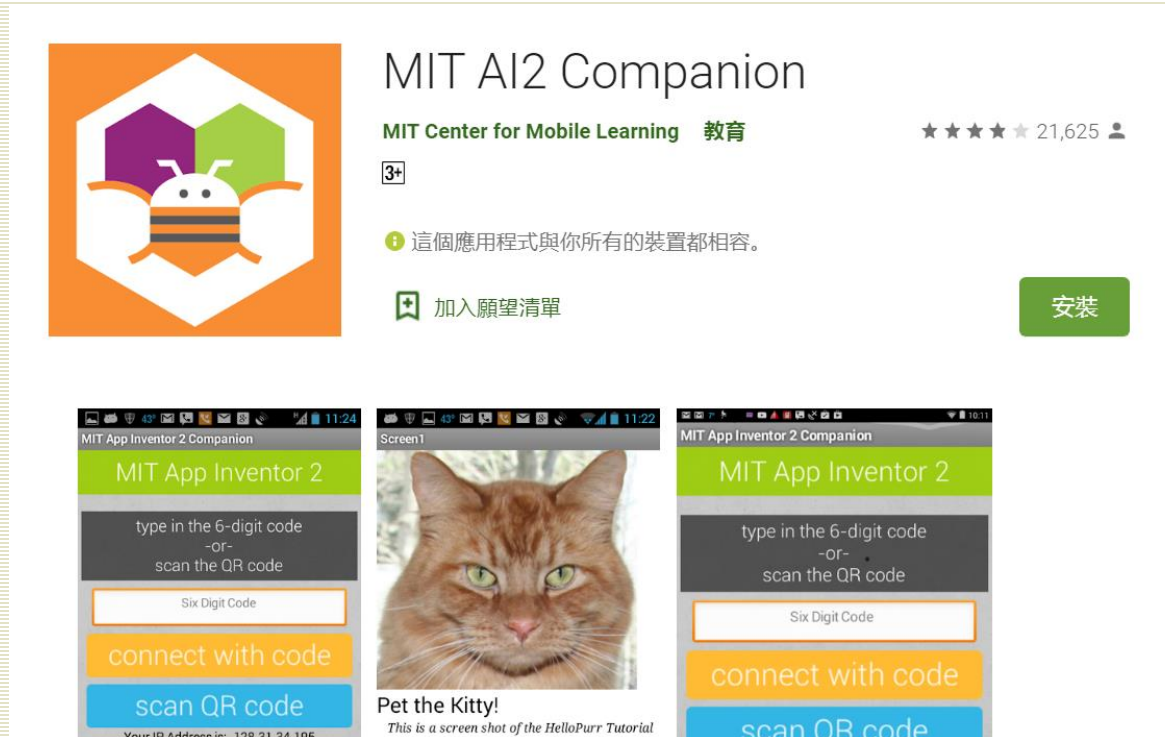
- 在手機上安裝App Inventor Companion（簡稱AI Companion，透過WiFi與App Inventor開發環境連線。
- 在電腦上使用軟體模擬器測試
- 透過電腦USB與手機連線





MIT AI2 Companion

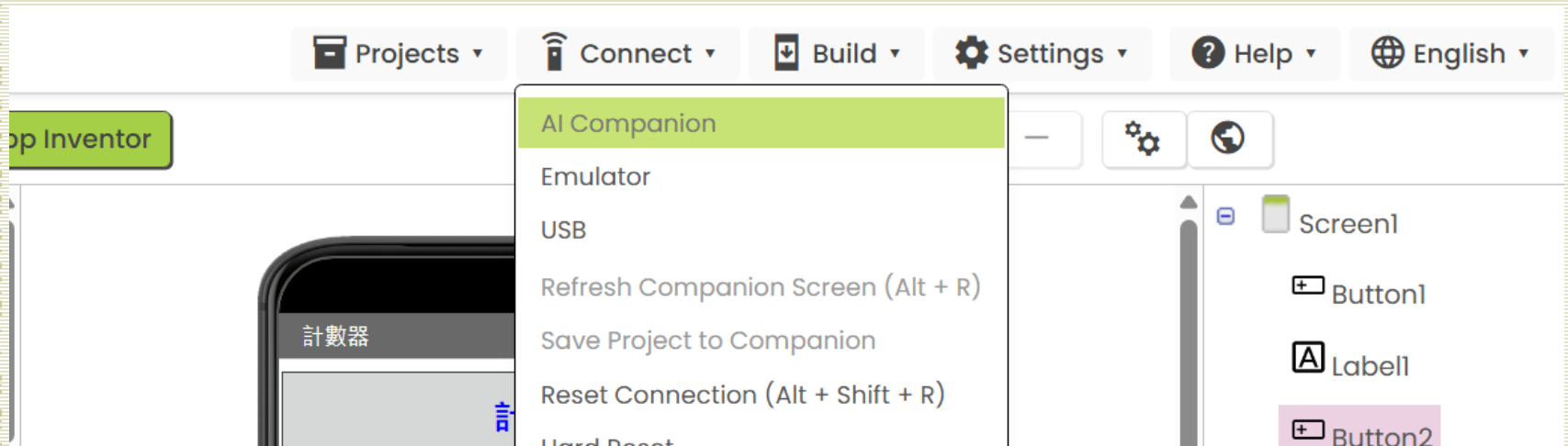
- ◆ 使用 Android 手機
- ◆ 打開 Google Play 搜尋 MIT AI2 ...





MIT AI2 Companion

◆ 使用相同 WIFI 分享器





產生 QR-code

Connect to Companion



Your
code is:

ssohjl

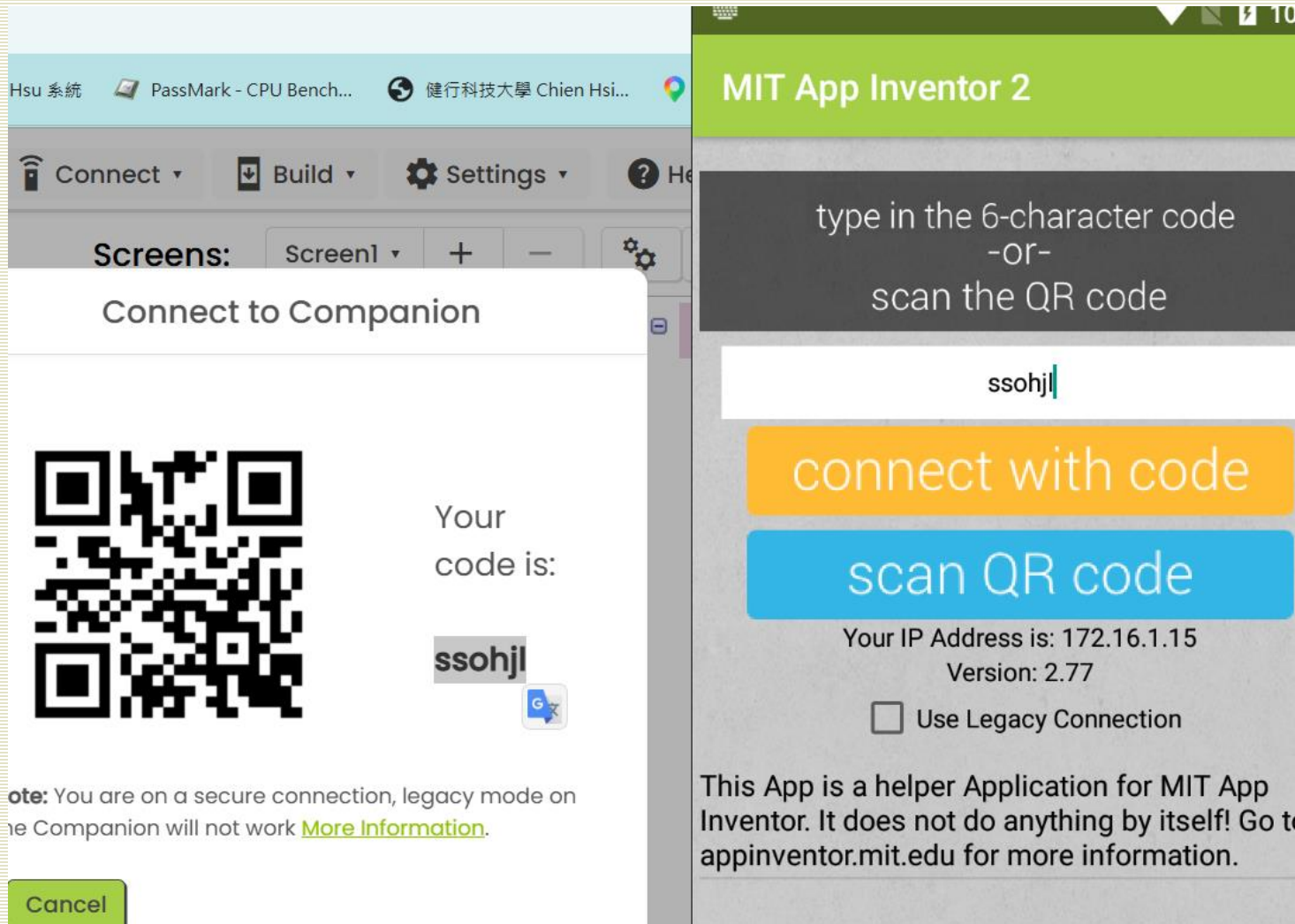
Note: You are on a secure connection, legacy mode on the Companion will not work [More Information](#).

Cancel





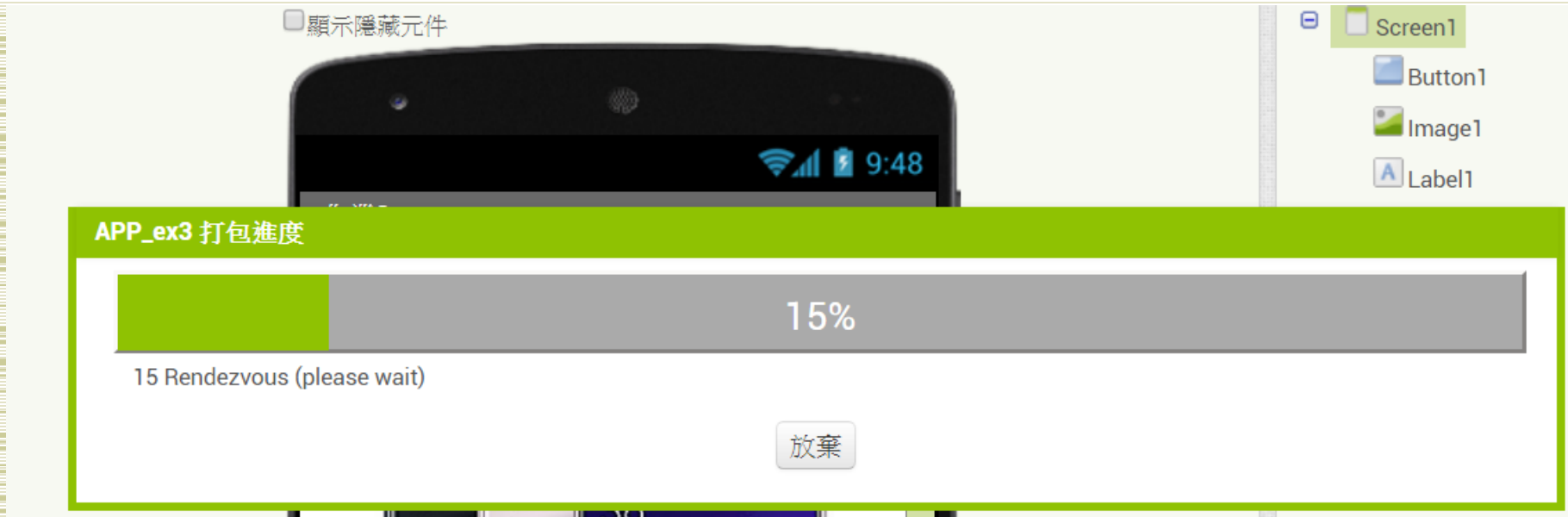
手機畫面





MIT AI2 Companion

- ◆ 使用 Scan QR code , 打包下載後WIFI連線





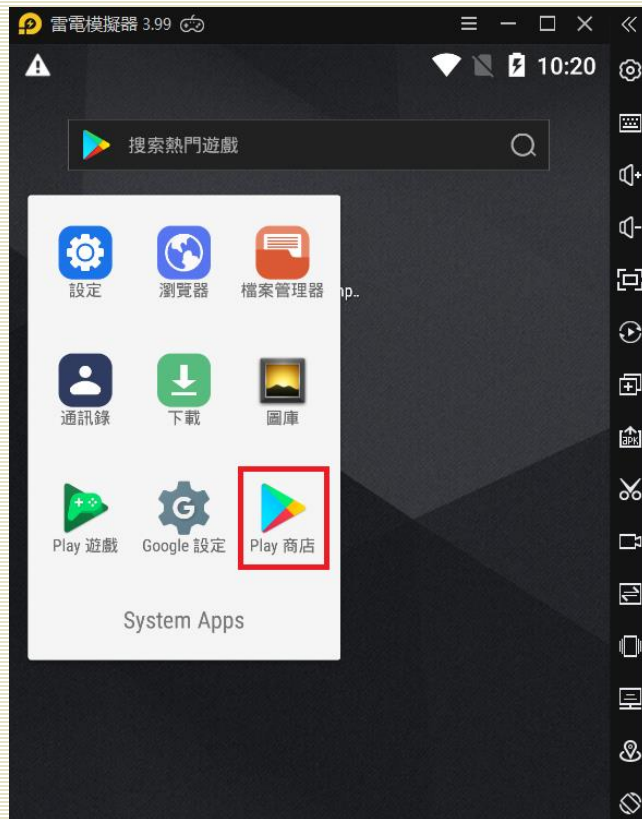
開啟 LDPlayer

- ◆ 啟動 google play
- ◆ 輸入 google帳號
- ◆ 安裝 MIT AI Companion (或使用 .apk)
- ◆ 啟動 MIT APP Inventor 2
 - 1.模擬器
 - 2.AI Companion
 - 3.產生 apk



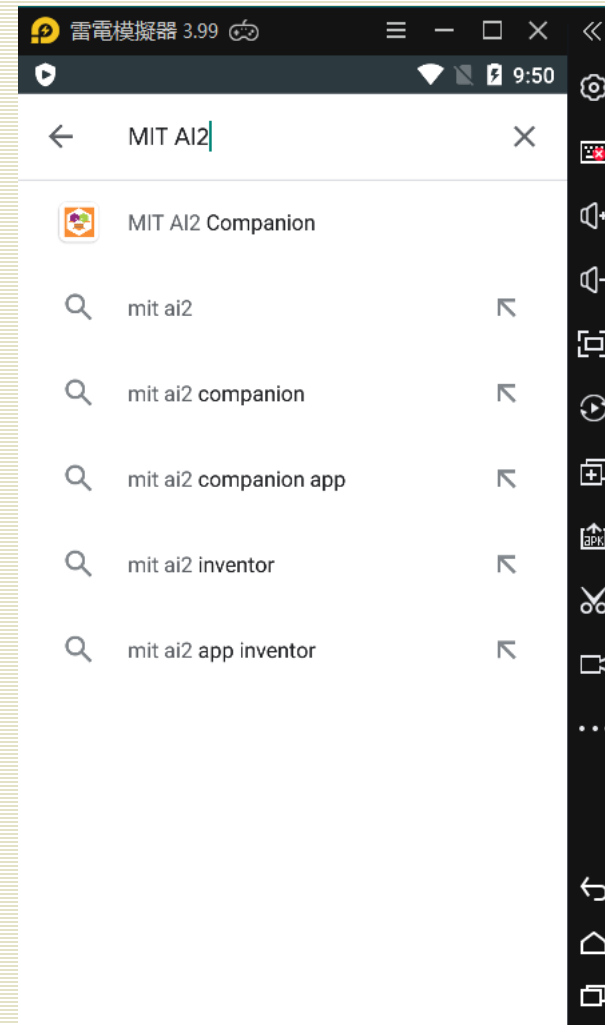


Play 商店



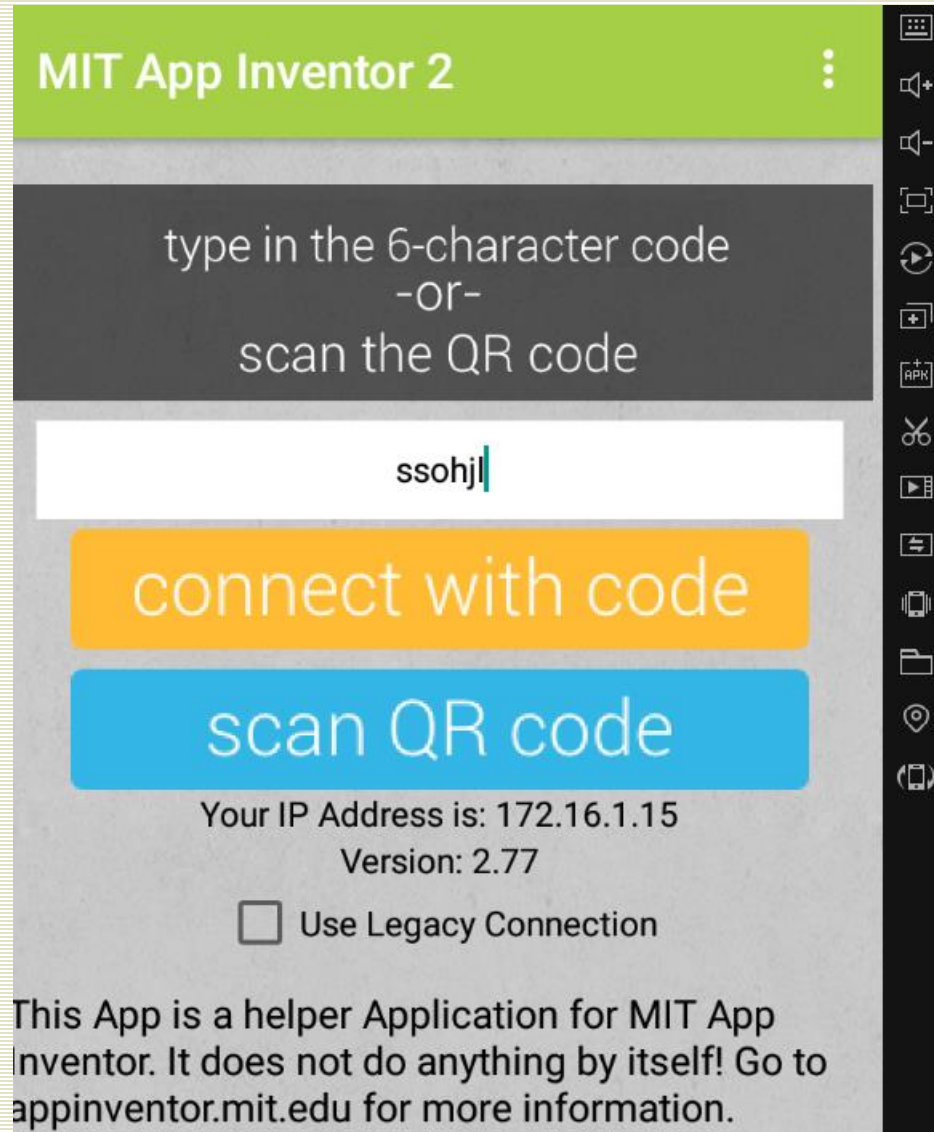


Play 商店





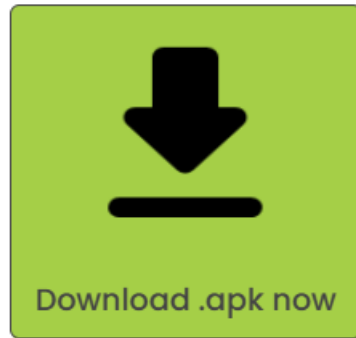
安裝 AI Companions





打包APK

Android App for Counter11



Click the button to download the app, right-click on it to copy a download link, or scan the code with a barcode scanner to install.

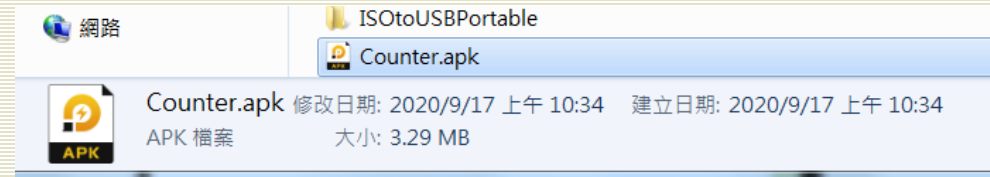
Note: this link and barcode are only valid for 2 hours. See [the FAQ](#) for info on how to share your app with others.

Dismiss





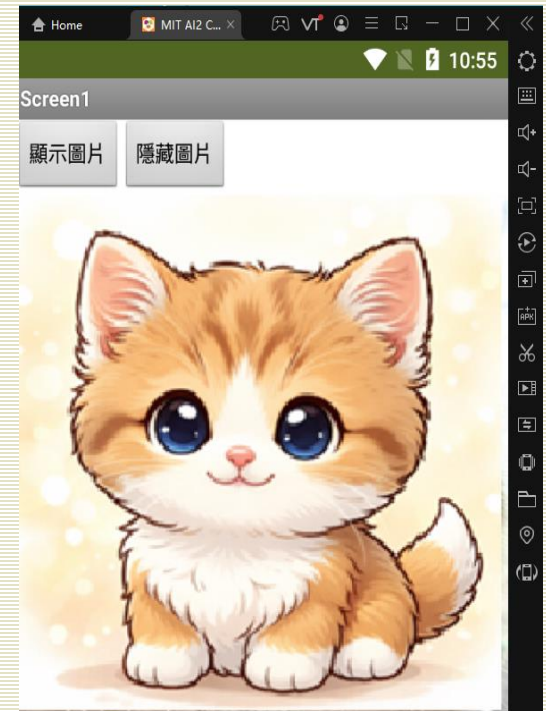
Counter.apk 拖拉到 LDplayer





實作範例: Hello

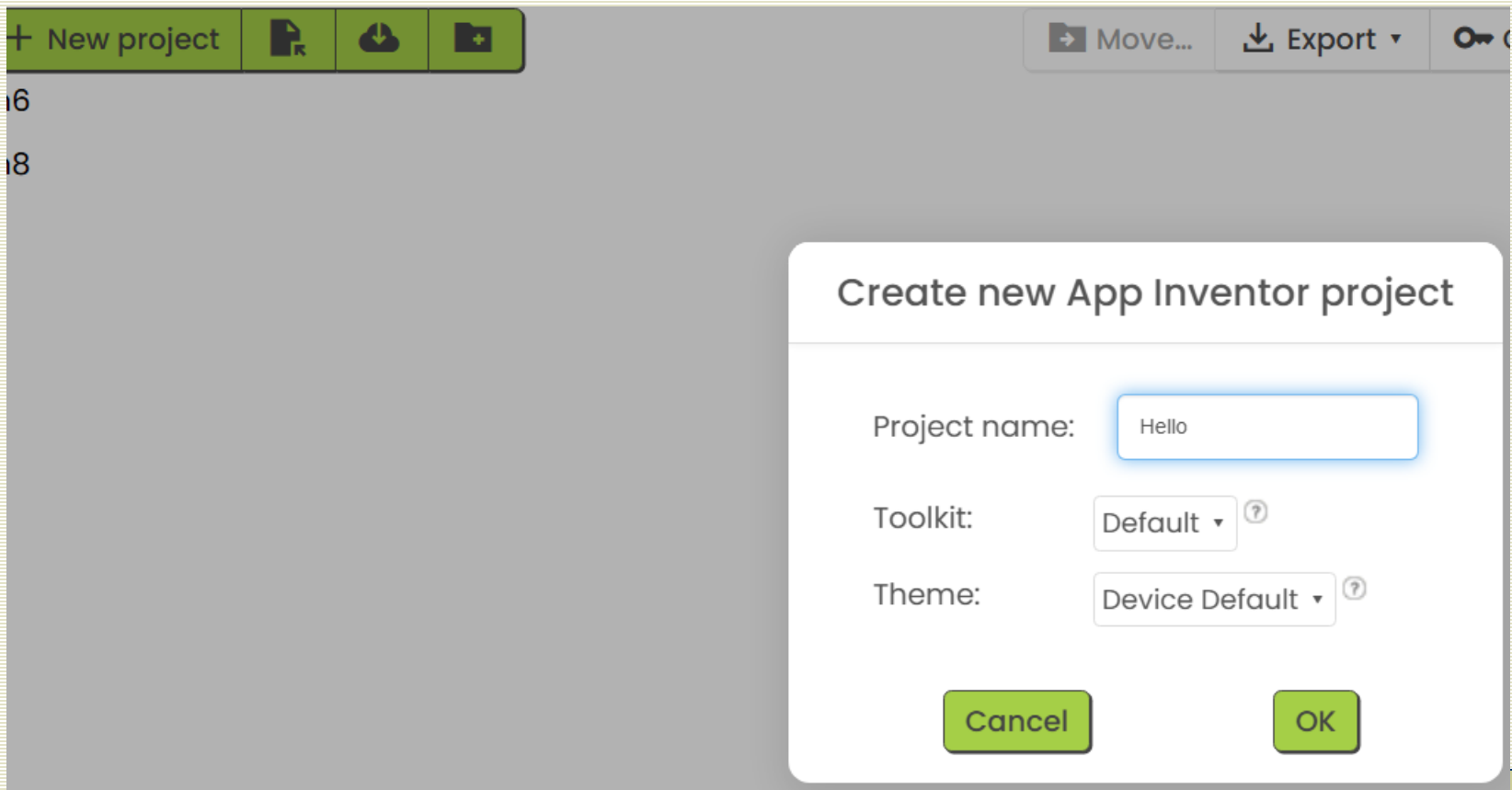
- ◆ 二個 Button
- ◆ 一個貓貓影像
(Kitty.png)
- ◆ 按顯示影像:
貓貓影像出現
- ◆ 按隱藏影像:
貓貓影像消失





產生新專案

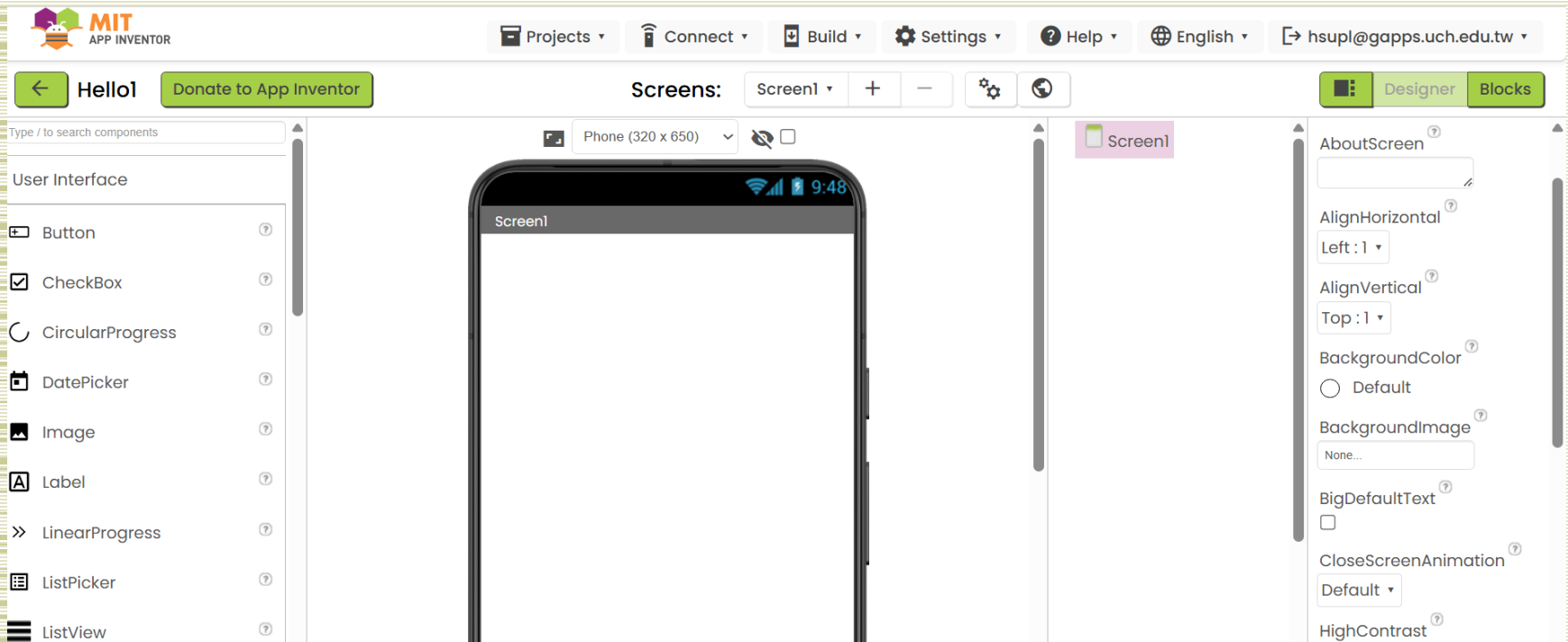
➤ 點 New project --> Hello





App Inventor Designer (版面佈置)

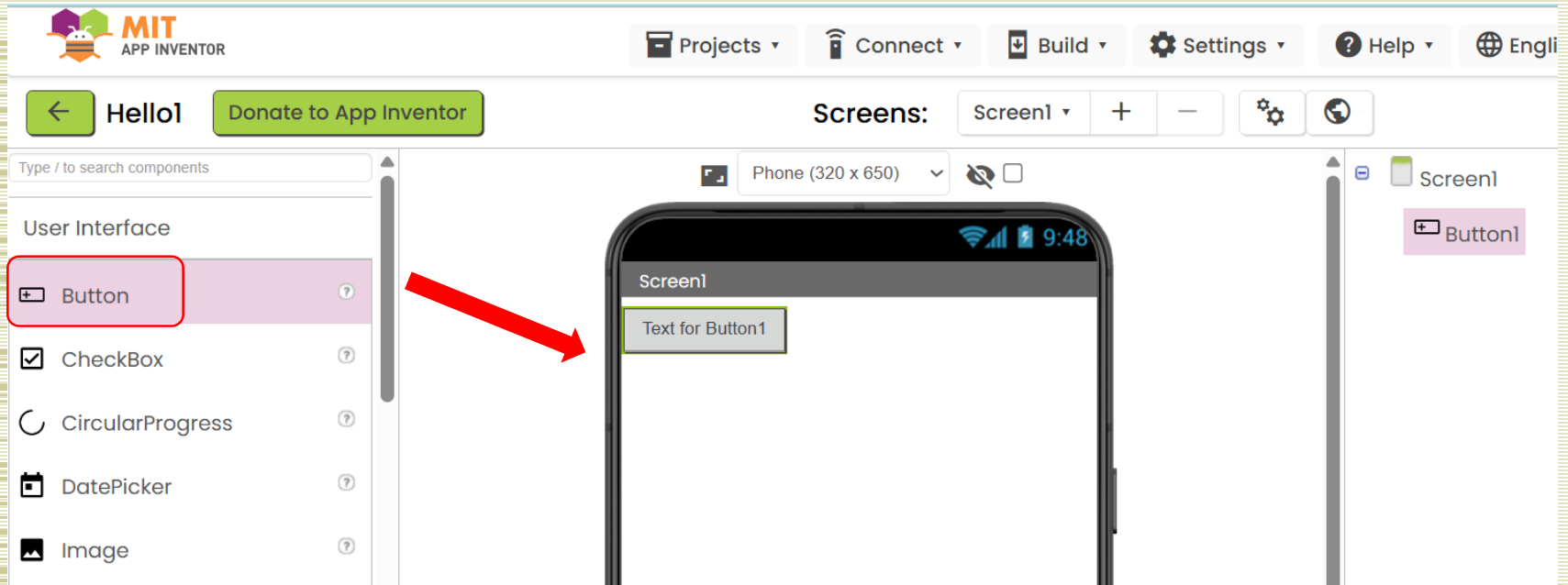
➤ 產生空白的 Hello Screen1 畫面:





App Inventor Designer (版面佈置)

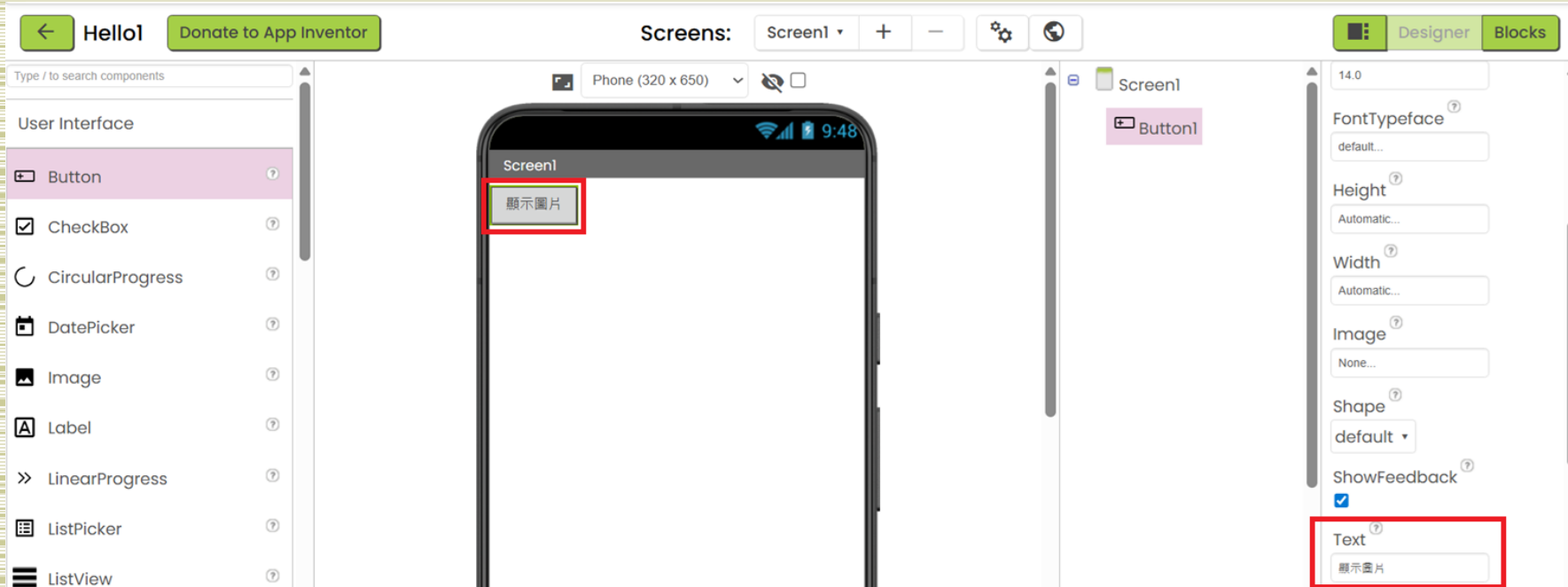
👉 從 Palette 拉入 第一個 Button,





App Inventor Designer (版面佈置)

👉 在 **Properties** 的 **Text**，改文字：**顯示圖片**





App Inventor Designer (版面佈置)

- ◆ 再拉入 第二個Button, 改 Text: 隱藏圖片

The screenshot illustrates the App Inventor Designer interface. On the left, the '元件面板' (Component Palette) is visible, showing a list of components under '使用者介面' (User Interface). The '按鈕' (Button) component is highlighted, and a red arrow points from it to the '隱藏圖片' (Hide Image) button in the 'Screen1' view. The '隱藏圖片' button is highlighted with a red box. On the right, the 'Properties' panel shows the 'Button1' properties, with the 'Text' property set to '隱藏圖片' and highlighted with a red box. The 'Image' panel shows the 'Image' property set to 'None...'.





App Inventor Designer (版面佈置)

👉 從 元件面板 → 界面配置 → 水平配置

The screenshot displays the App Inventor Designer interface. At the top, there's a green header bar with the text "Hello" and a dropdown menu set to "Screen1". Below the header, there are two main panels: "元件面板" (Component Panel) on the left and "工作面板" (Workspace) on the right.

In the "元件面板", the "介面配置" (Interface Configuration) section is expanded. The "水平配置" (Horizontal Layout) option is highlighted with a red box and a mouse cursor icon. A red lightning bolt arrow points from this option to the workspace.

The "工作面板" shows a mobile device screen. At the top, there's a status bar with icons for Wi-Fi, signal strength, battery, and the time "9:48". Below the status bar, the screen is titled "Screen1". There are two buttons: "顯示圖片" (Show Image) and "隱藏圖片" (Hide Image). The "隱藏圖片" button is highlighted with a green box. Below the buttons, there's a large white rectangular area, which is also highlighted with a red box.





App Inventor Designer (版面佈置)

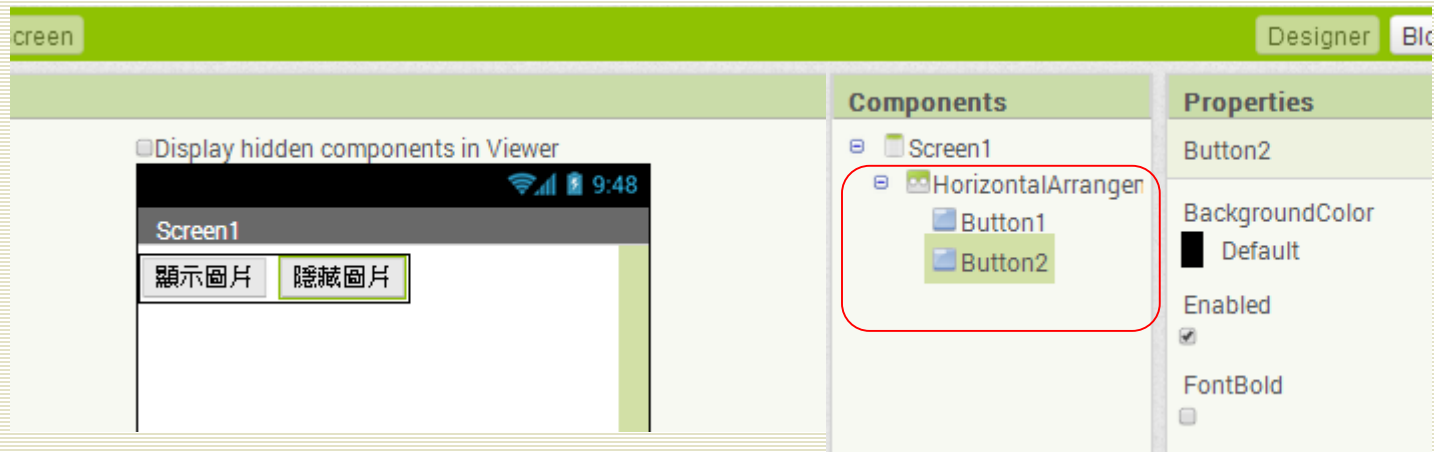
- ◆ 從 layout 拉入 水平配置 HorizotalArrangement





App Inventor Designer (版面佈置)

- ◆ 將兩個Button 放入 HorizontalArrangement





App Inventor Designer (版面佈置)

- ◆ 從 Palette 的 User Interface 拉入 Image,

The screenshot illustrates the App Inventor Designer interface. On the left, the 'Hello' palette is visible, with the 'User Interface' section expanded. The 'Image' component is highlighted, and a red arrow points to it. In the center, the 'Screen1' view is shown, with a red box highlighting the 'Image' icon in the top-left corner. On the right, the 'Components' panel shows the hierarchy: 'Screen1' contains 'HorizontalArranger', which contains 'Button1', 'Button2', and 'Image1'. The 'Properties' panel on the far right shows the properties for 'Image1', including 'Picture' (set to 'None...'), 'Visible' (set to 'showing'), 'Width' (set to 'Automatic...'), and 'Height' (set to 'Automatic...'). The 'Media' panel at the bottom right has an 'Upload File ...' button.





App Inventor Designer (版面佈置)

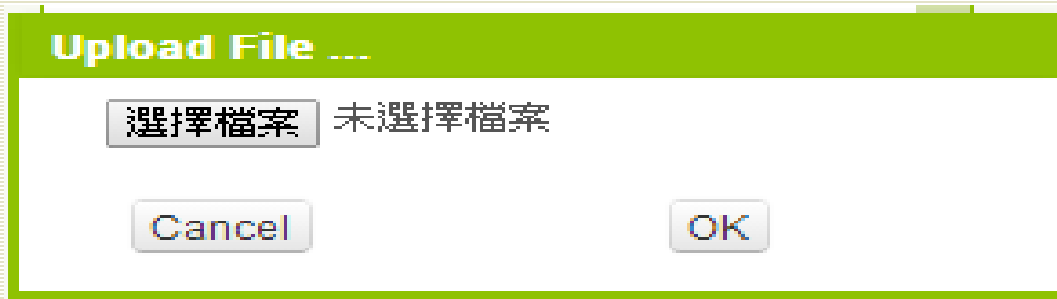
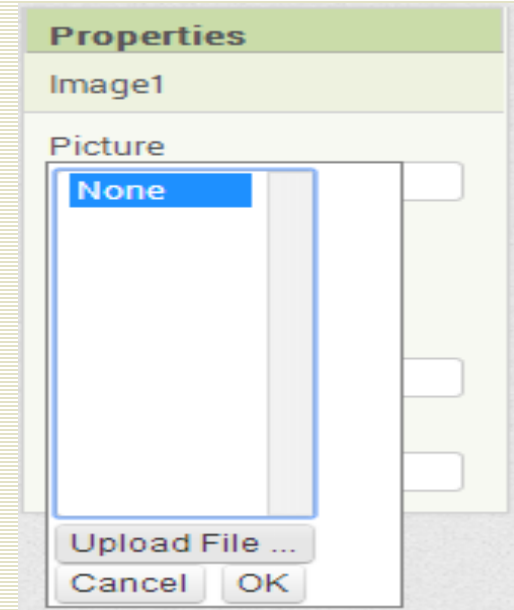


Picture

- ✓ 點 None
- ✓ 點 Upload, 準備加入圖片!



點選檔案:

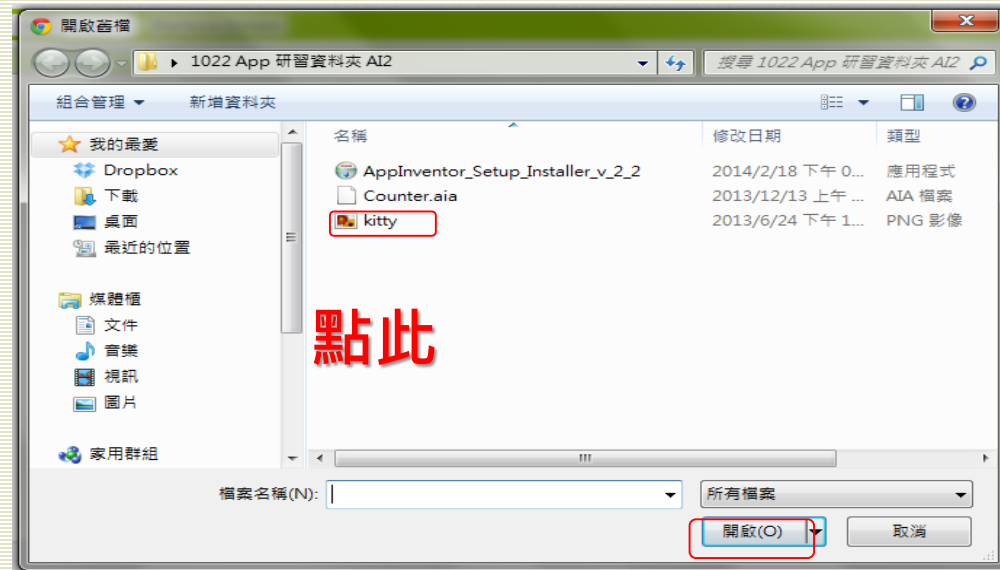




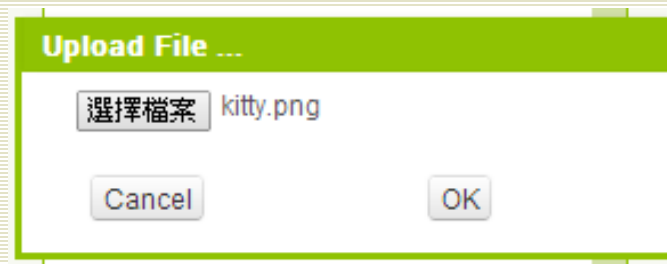
App Inventor Designer (版面佈置)

加入圖片檔

Kitty.png



按 Ok





App Inventor Designer (版面佈置)

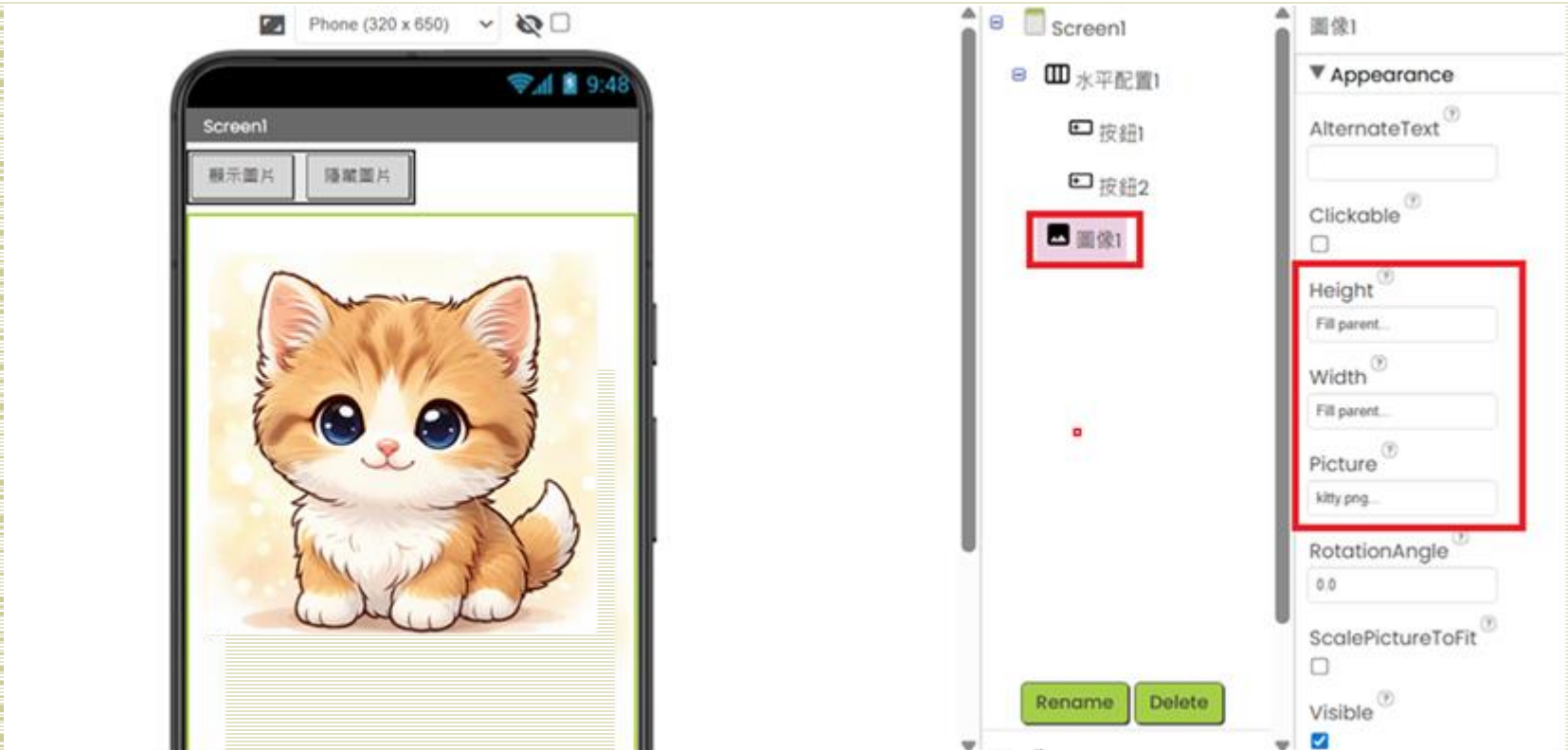
➤ 完成後版面佈置畫面：





App Inventor Designer (版面佈置)

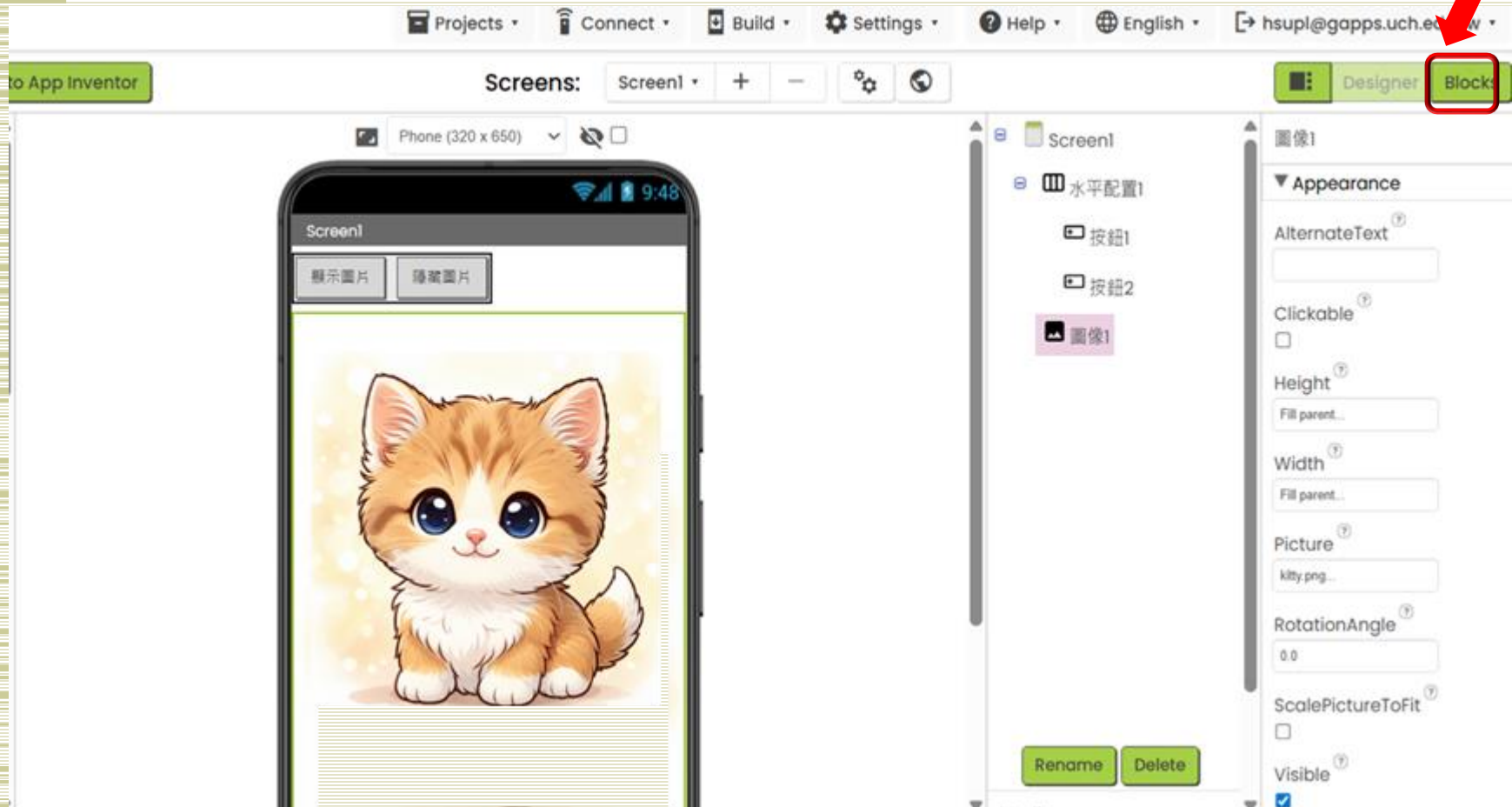
➤ 完成後版面佈置畫面：





Block Editor (堆程式)

按 Blocks 進入 拼圖編輯器 --> 寫程式 按此





Block Editor (堆程式)

- ◆ 從 Screen1 的 按鈕1 及 按鈕2 取入:





Block Editor (堆程式)

◆ 再從 Image1 取入可見性 visible 屬性

Block Editor (堆程式) 截圖：

畫面標題：Hello

Screen1 新增螢幕 刪除螢幕

方塊 (Block) 列表：

- 內件方塊 (Component Block)
 - 流程控制 (Flow Control)
 - 邏輯 (Logic)
 - 數學 (Math)
 - 文字 (Text)
 - 清單 (List)
 - Dictionaries
 - 顏色 (Color)
 - 變數 (Variable)
 - 程序 (Script)
- Screen1
 - 水平配置1 (Horizontal Layout 1)
 - 按鈕1 (Button 1)
 - 按鈕2 (Button 2)
 - 圖像1 (Image 1)** (被紅框圈出)
- 任意元件 (Any Component)

工作面板 (Workspace) 內容：

- 設 圖像1 . 旋轉角度 為
- 執行
- 設 圖像1 . 放大/縮小圖片來適應尺寸 為
- 執行
- 圖像1 . 縮放模式
- 執行
- 設 圖像1 . 縮放模式 為
- 圖像1 . 可見性
- 設 圖像1 . 可見性 為** (被紅框圈出)
- Specifies whether the Image Value is true if the Image is
- 圖像1 . 寬度
- 設 圖像1 . 寬度 為
- 設 圖像1 . 寬度百分比 為
- 顯示警告
- 圖像1

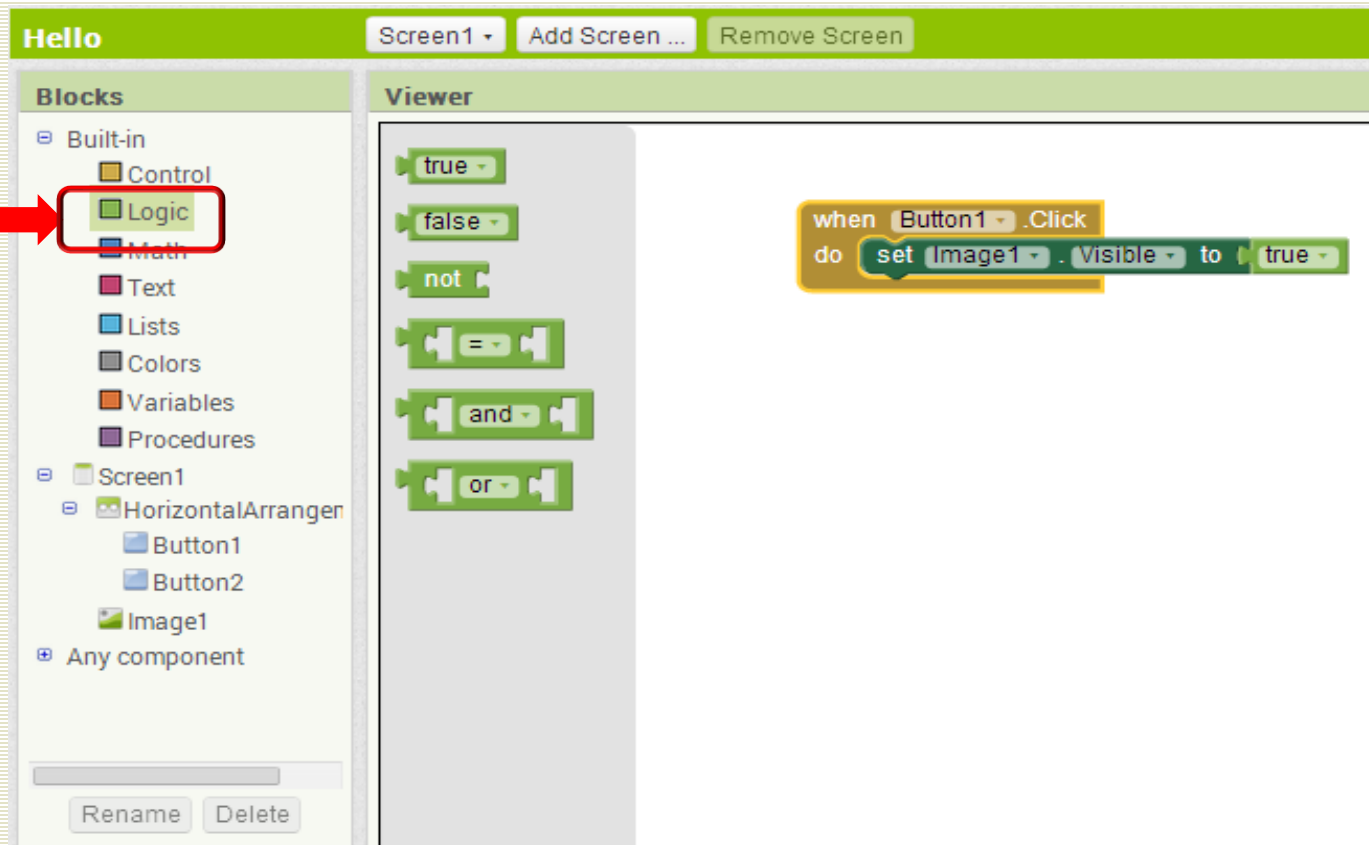




Block Editor (堆程式)

- ◆ 從 Built-In 的 Logic(邏輯) 取入: true (真)

按此 →





Block Editor (堆程式)

- ◆ 同理, 對 按鈕2 堆程式, 但可見性 to 假 **false**:

The screenshot shows the 'Hello' screen in a block editor. The top bar has 'Screen1' and buttons for '新增螢幕' (Add Screen) and '刪除螢幕' (Delete Screen). The left sidebar, titled '方塊' (Blocks), lists categories: 內件方塊 (UI Widgets), 流程控制 (Flow Control), 邏輯 (Logic), 數學 (Math), 文字 (Text), 清單 (List), Dictionaries, 顏色 (Color), 變數 (Variable), and 程序 (Script). The main workspace, titled '工作面板' (Workspace), contains two scripts. The first script is triggered by '當 按鈕1 被點選' (When Button 1 is clicked) and contains a '執行 設 圖像1 可見性 為 真' (Execute Set Image 1 visibility to True) block. The second script is triggered by '當 按鈕2 被點選' (When Button 2 is clicked) and contains a '執行 設 圖像1 可見性 為 假' (Execute Set Image 1 visibility to False) block.

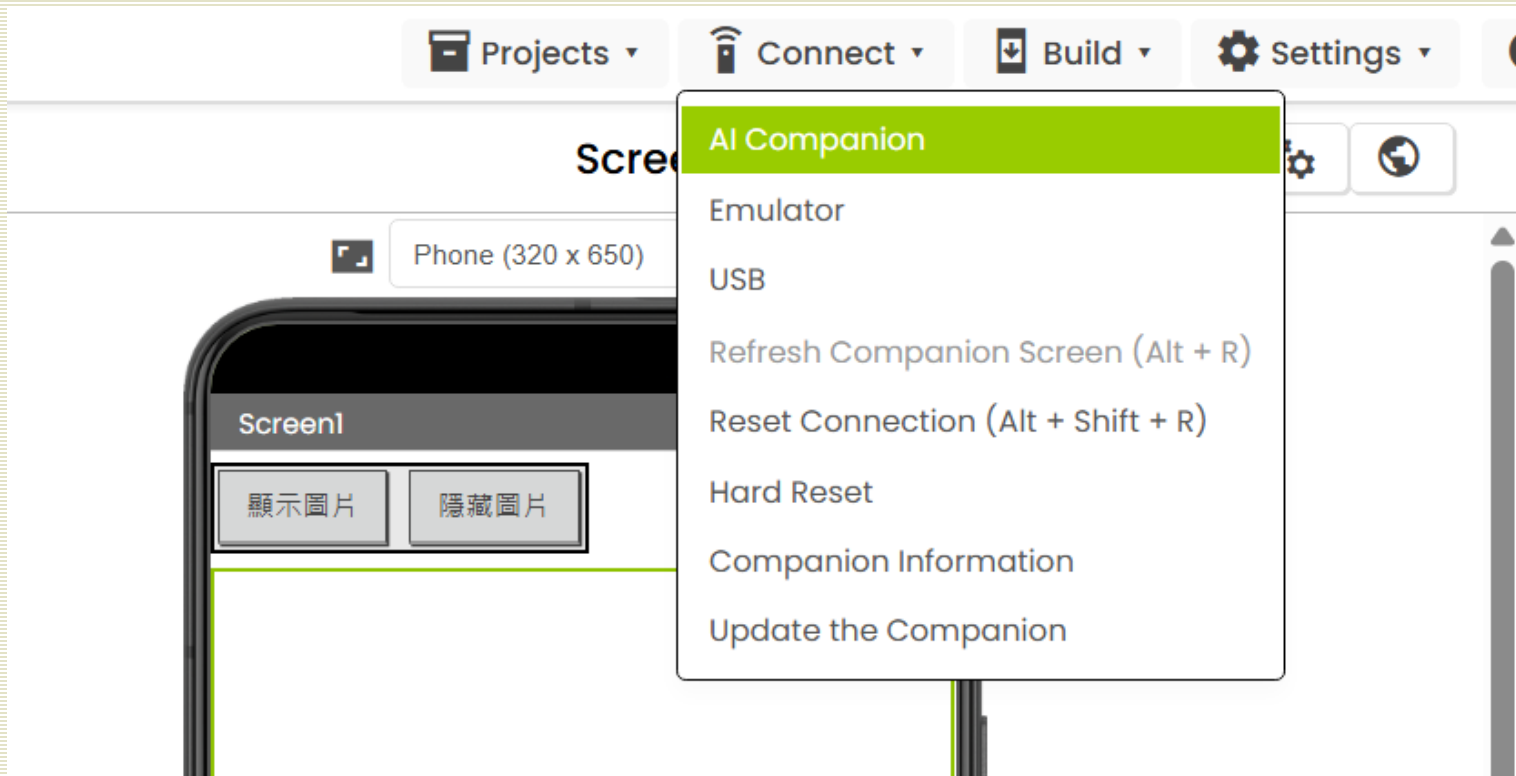




AI Companion

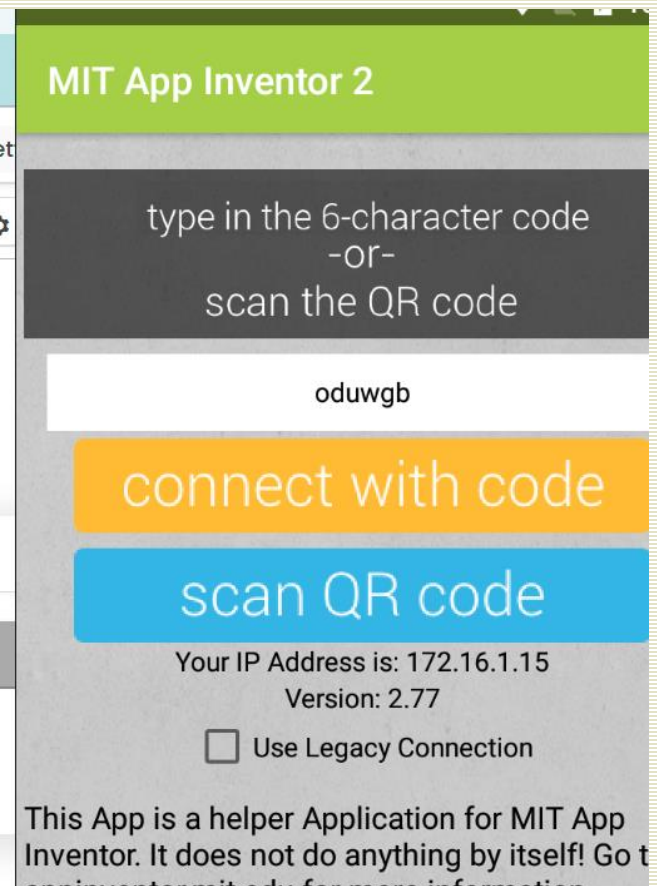
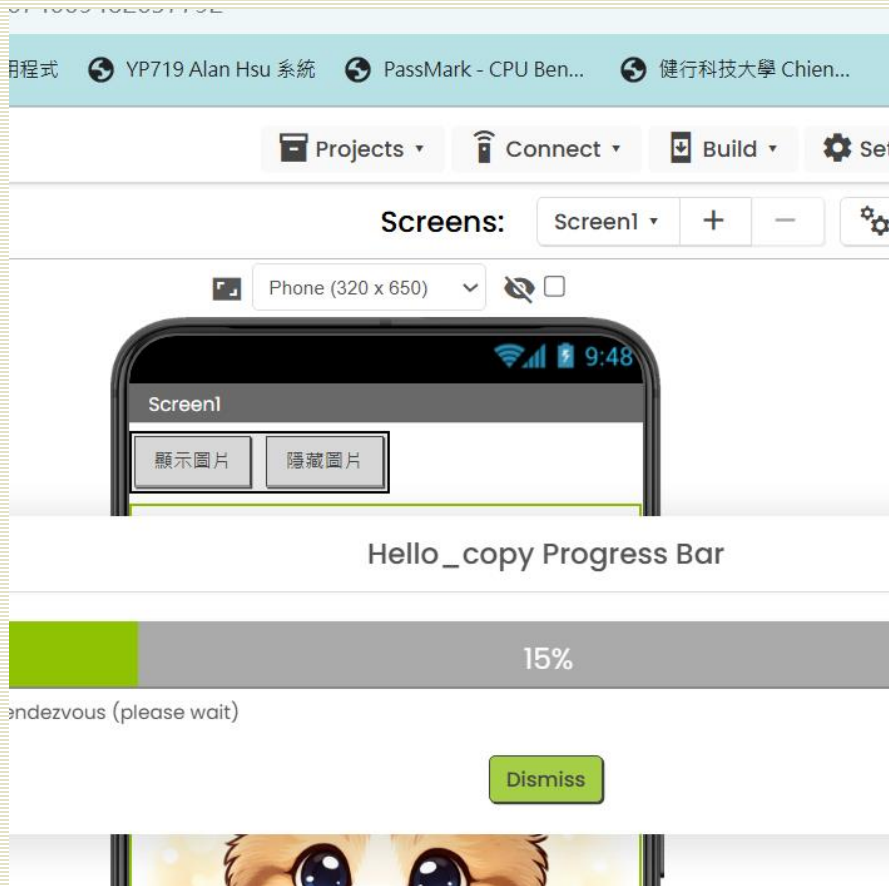


Connect -> AI Companion





AI Companion





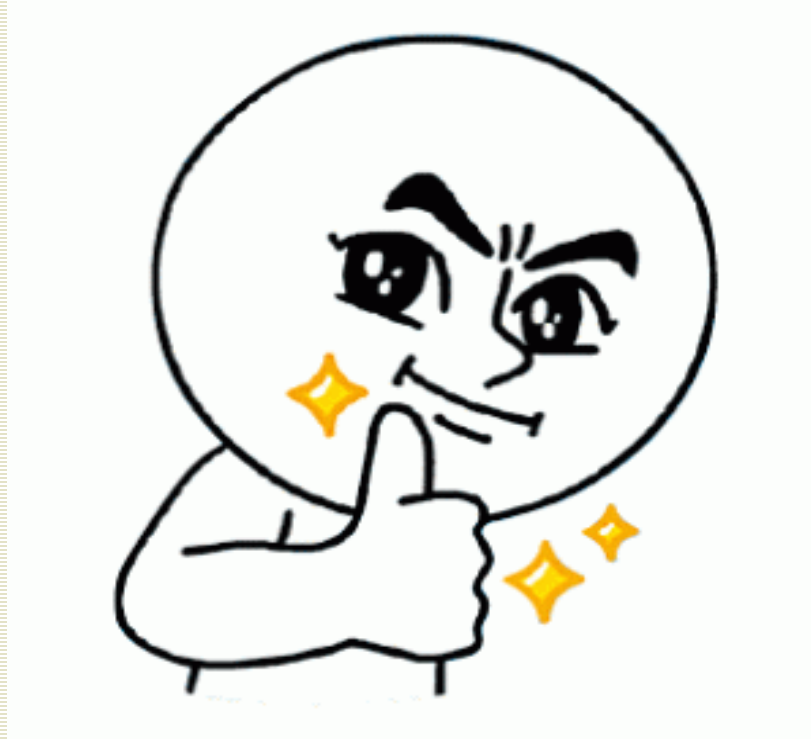
Emulator (模擬執行)

- ◆ 完成了!!
- ◆ 按顯示影像:
貓貓影像出現
- ◆ 按隱藏影像:
貓貓影像消失





實 作 討 論





練習：HelloTest

- 一開始:

貓貓影像出現

你好!我是可愛小貓

- 按下請按我:

小狗狗出現了!

您好!我是可愛小狗狗





產生新專案

➤ 點 專案 --> 新增專案 (HelloTest)





App Inventor Designer (畫面編排)

➤ 產生空白的 HelloTest Screen1 畫面:

The screenshot displays the MIT App Inventor Designer web interface. At the top, the MIT App Inventor logo is on the left, and navigation links (專案, 連線, 打包apk, Settings, 幫助) are in the center. On the right, there are links for '我的專案', 'View Trash', '指南', '回報問題', '繁體中文', and a user email 'Alanhh318@gmail.com'. Below the navigation bar, a green header bar contains the project name 'HelloTest', a dropdown menu for 'Screen1', and buttons for '新增螢幕', '刪除螢幕', and '發佈作品到Gallery'. On the far right of this bar are two tabs: '畫面編排' (selected) and '程式設計'.

The main workspace is divided into four panels:

- 元件面板 (Component Palette):** Located on the left, it contains a search bar and a list of UI components under the heading '使用者介面'. The components include: 按鈕, 複選盒, 日期選擇器, 圖像, 標籤, 清單選擇器, 清單顯示器, 對話框, 密碼輸入盒, 滑桿, 下拉式選單, Switch, 文字輸入盒, 時間選擇器, and 網路瀏覽器. Below this list are sections for '介面配置' and '多媒體'.
- 工作面板 (Workspace):** The central area, highlighted with a red box. It shows a mobile device screen with a status bar at the top displaying 'Screen1', signal strength, battery, and the time '9:48'. The screen itself is blank white. Above the device, there is a checkbox for '顯示隱藏元件' and a dropdown for '手機尺寸 (505,320)'.
- 元件清單 (Component List):** Located on the right side of the workspace, it shows a single component 'Screen1' with a small square icon next to it. Below the list are buttons for '重新命名' and '刪除'.
- 元件屬性 (Component Properties):** The rightmost panel, which lists various properties for the selected 'Screen1' component. These include: '應用說明' (Application Description), '突顯顏色' (Highlight Color) with a '預設' (Default) option, '水平對齊' (Horizontal Alignment) set to '靠左: 1', '垂直對齊' (Vertical Alignment) set to '靠上: 1', 'App名稱' (App Name) set to 'HelloTest', '背景顏色' (Background Color) with a '預設' option, '背景圖片' (Background Image) set to '無...', 'BlocksToolkit' set to 'All', '關閉螢幕動畫' (Close Screen Animation) set to '預設效果', '圖示' (Icon) set to '無...', and '開啟螢幕動畫' (Open Screen Animation) set to '預設效果'.





App Inventor Designer (畫面編排)

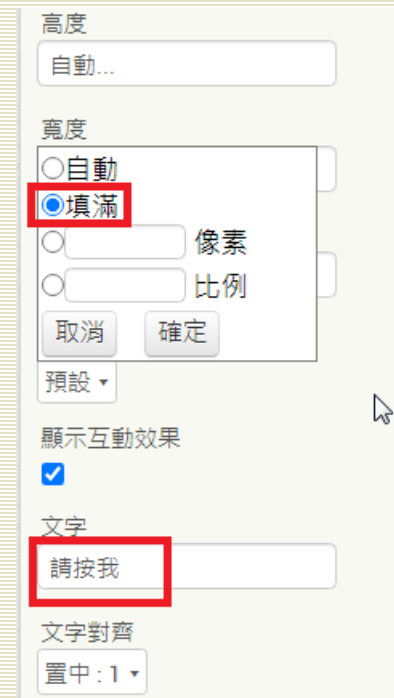
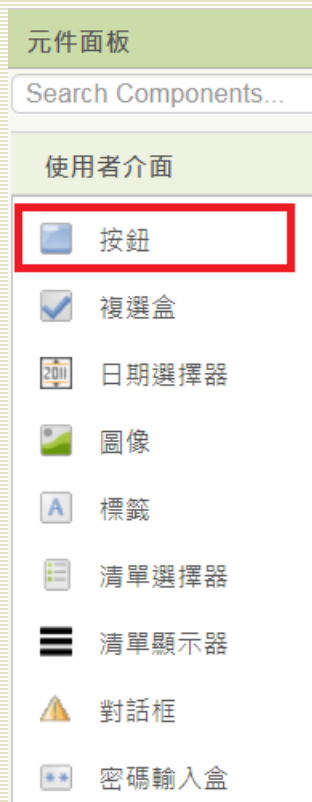
👉 從元件面板拉入第一個按鈕1,





App Inventor Designer (畫面編排)

👉 Text : 『請按我』, 並將Width 選為: Fill parent ...





App Inventor Designer (畫面編排)

◆ 從元件面板的拉入圖像,

The screenshot displays the App Inventor Designer interface with four main panels:

- 元件面板 (Component Palette):** Located on the left, it contains a search bar and a list of components under the heading "使用者介面" (User Interface). The "圖像" (Image) component is highlighted with a red box. A red arrow points from this box to the central canvas.
- Canvas:** The central workspace shows a mobile app screen titled "Screen1" with a status bar at the top displaying signal, battery, and time (9:48). Below the status bar is a button labeled "請按我" (Click me). A small image icon in the top-left corner of the screen is highlighted with a red box.
- 元件清單 (Component List):** Located on the right, it shows a tree view of the components on the screen: "Screen1", "按鈕1" (Button1), and "圖像1" (Image1). The "圖像1" component is highlighted with a green box.
- 元件屬性 (Component Properties):** Located on the far right, it shows the properties for the selected "圖像1" component. The properties include: "Clickable" (unchecked), "高度" (Height) set to "自動..." (Automatic), "寬度" (Width) set to "自動..." (Automatic), "圖片" (Image) set to "無..." (None), "旋轉角度" (Rotation angle) set to "0.0", "放大/縮小圖片來適應尺寸" (Scale image to fit) (unchecked), and "可見性" (Visibility) (checked).





App Inventor Designer (畫面編排)



圖片

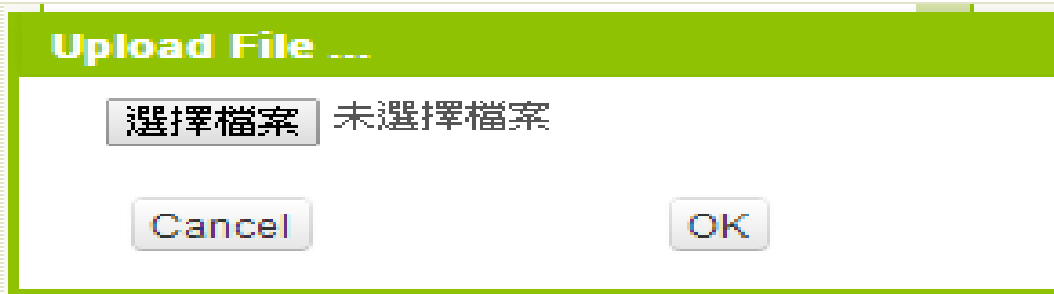
✓ 點 **Upload**, 準備加入圖片!

✓ 加入 kitty.png

✓ 加入 doggie.png



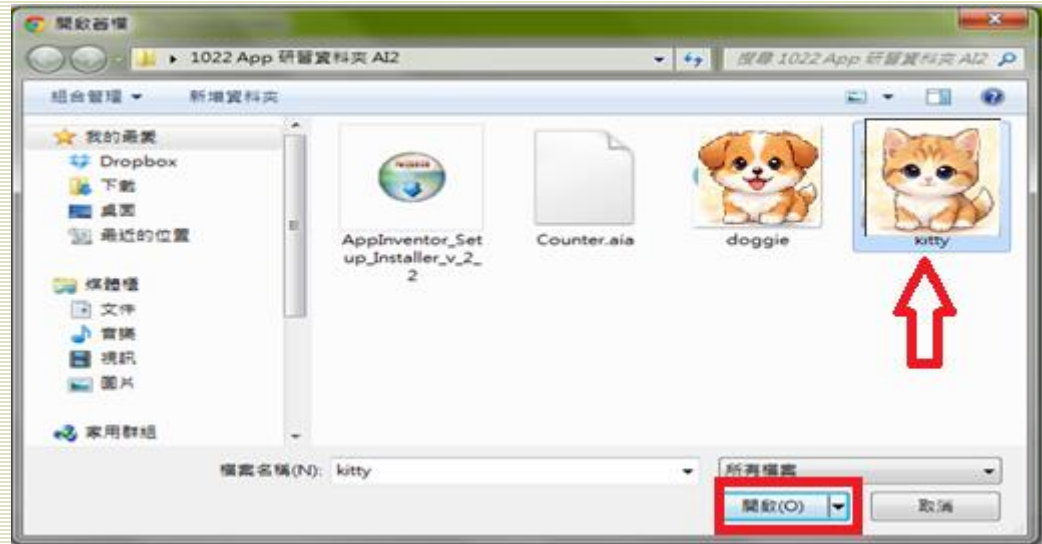
點選**選擇檔案**:



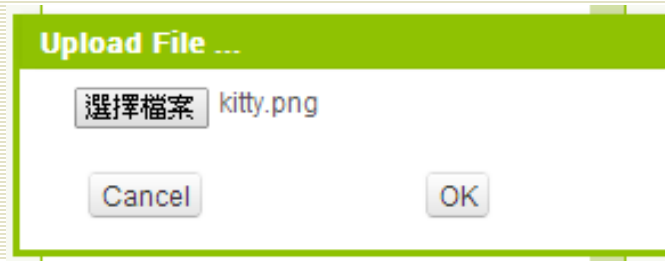


App Inventor Designer (畫面編排)

加入圖片檔
Kitty.png



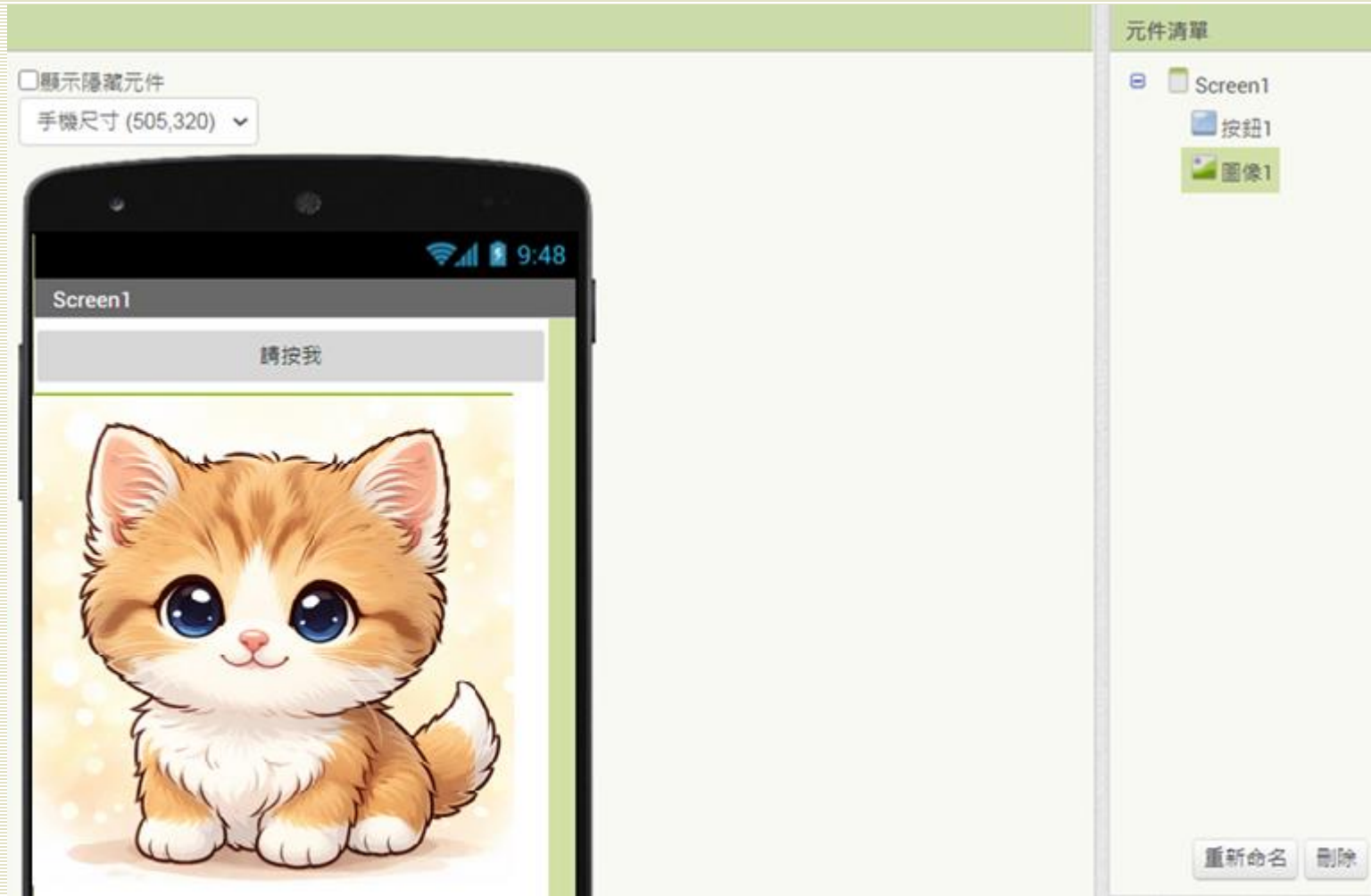
按 **Ok**





App Inventor Designer (畫面編排)

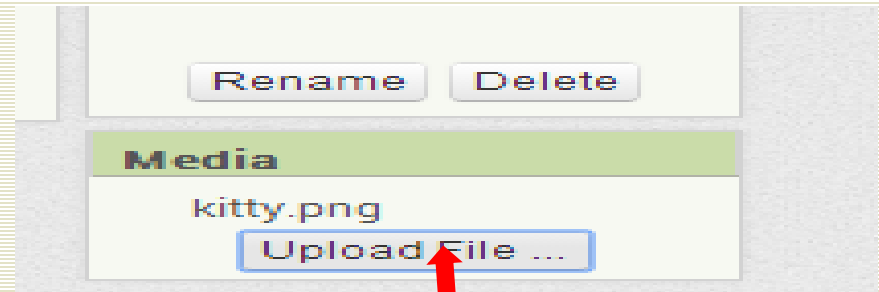
➤ 加入後畫面：





App Inventor Designer (畫面編排)

👉 再增加一個圖片：



點此

👉 點 選擇檔案

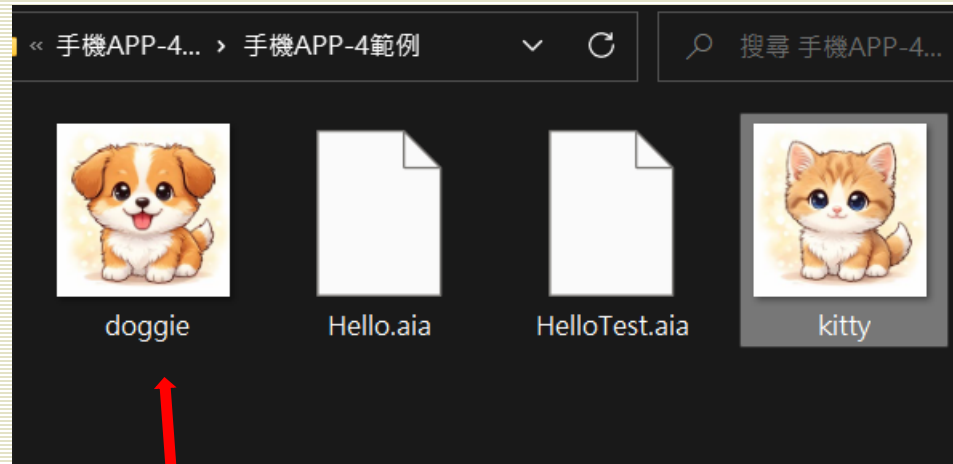
點此





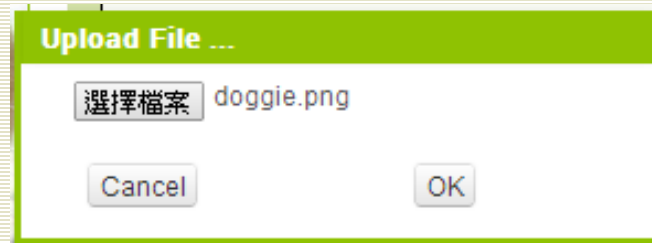
App Inventor Designer (畫面編排)

➡ 加入圖片檔
doggie.png



點此

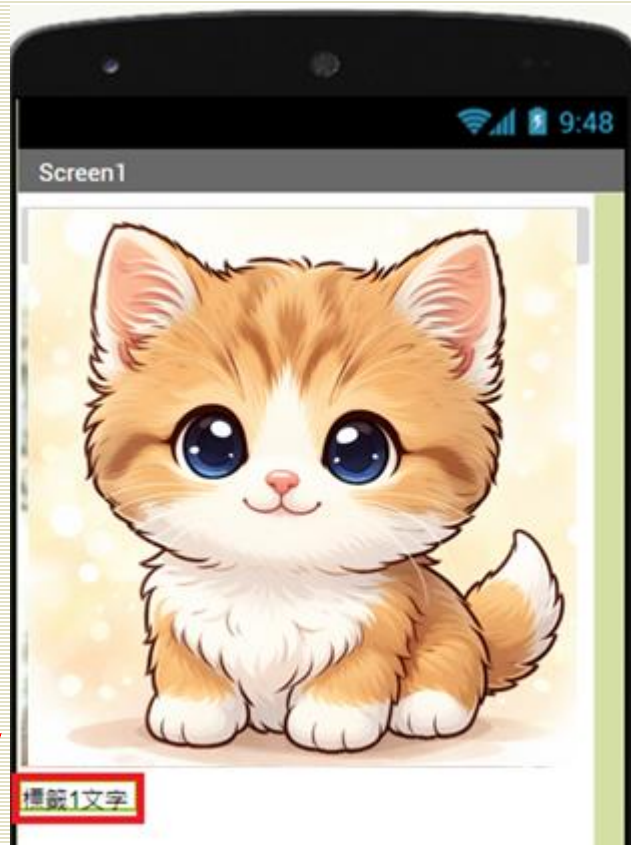
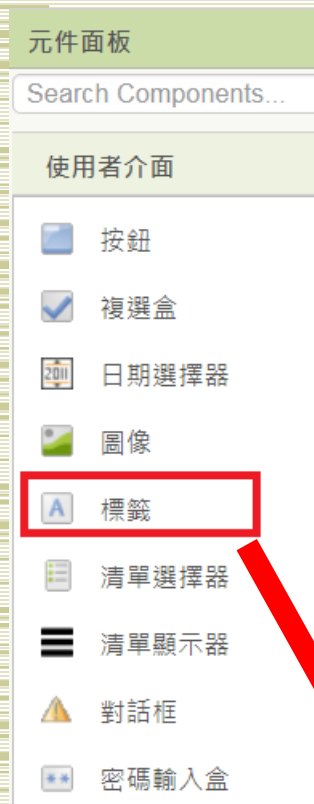
➡ 按 Ok





App Inventor Designer (畫面編排)

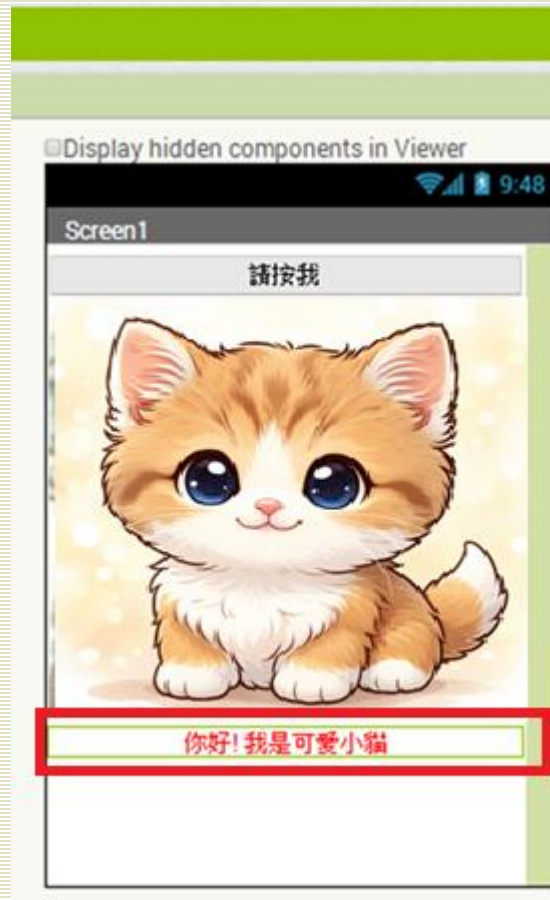
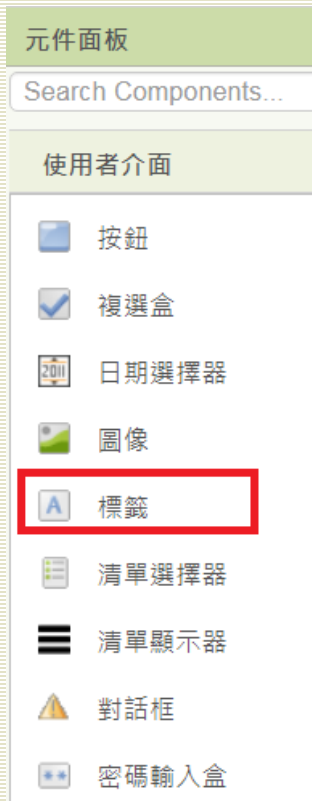
👉 將標籤拉入 Screen1,





App Inventor Designer (畫面編排)

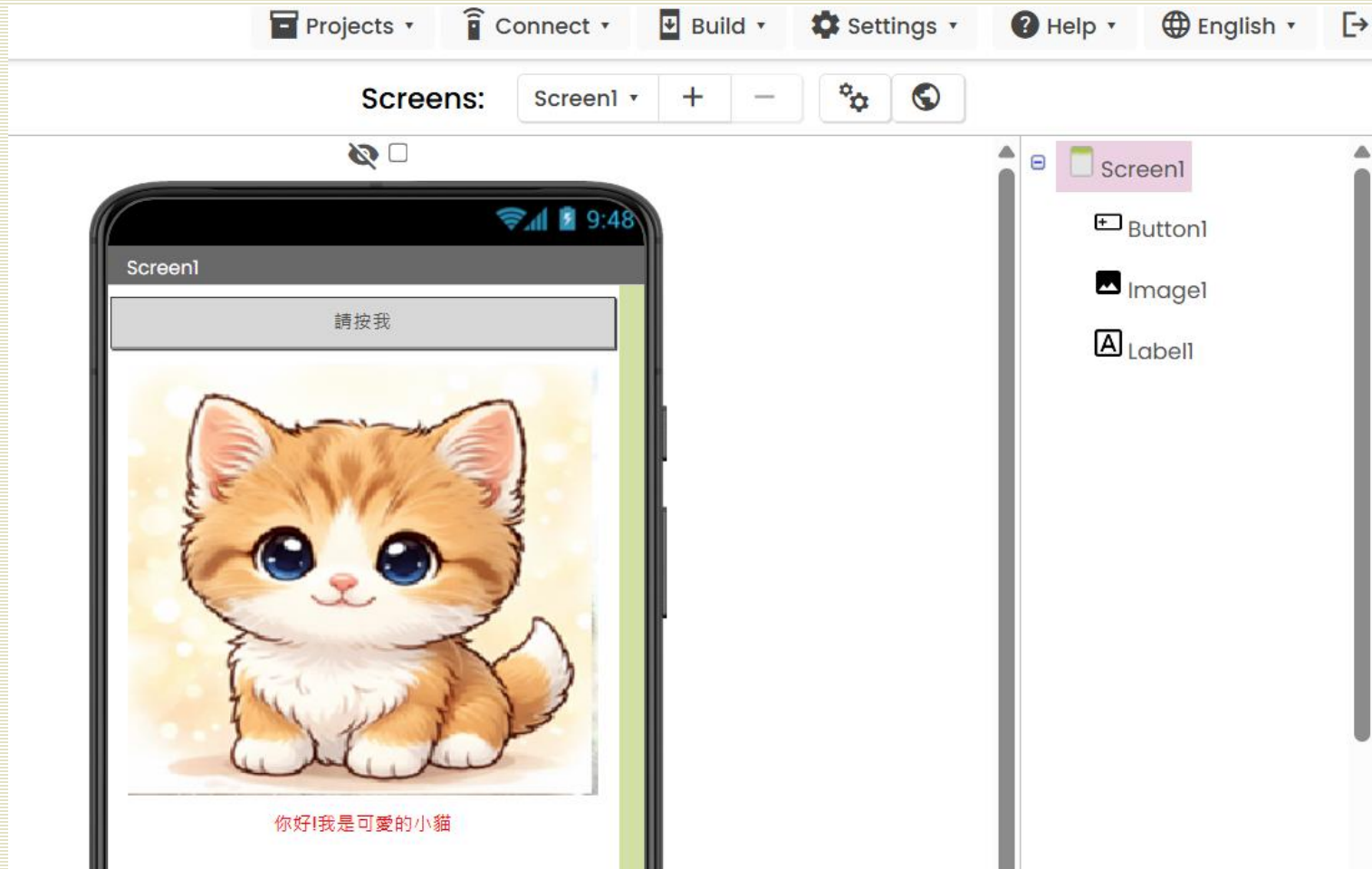
- ◆ 將文字內容改為: 『你好! 我是可愛小貓』





App Inventor Designer (畫面編排)

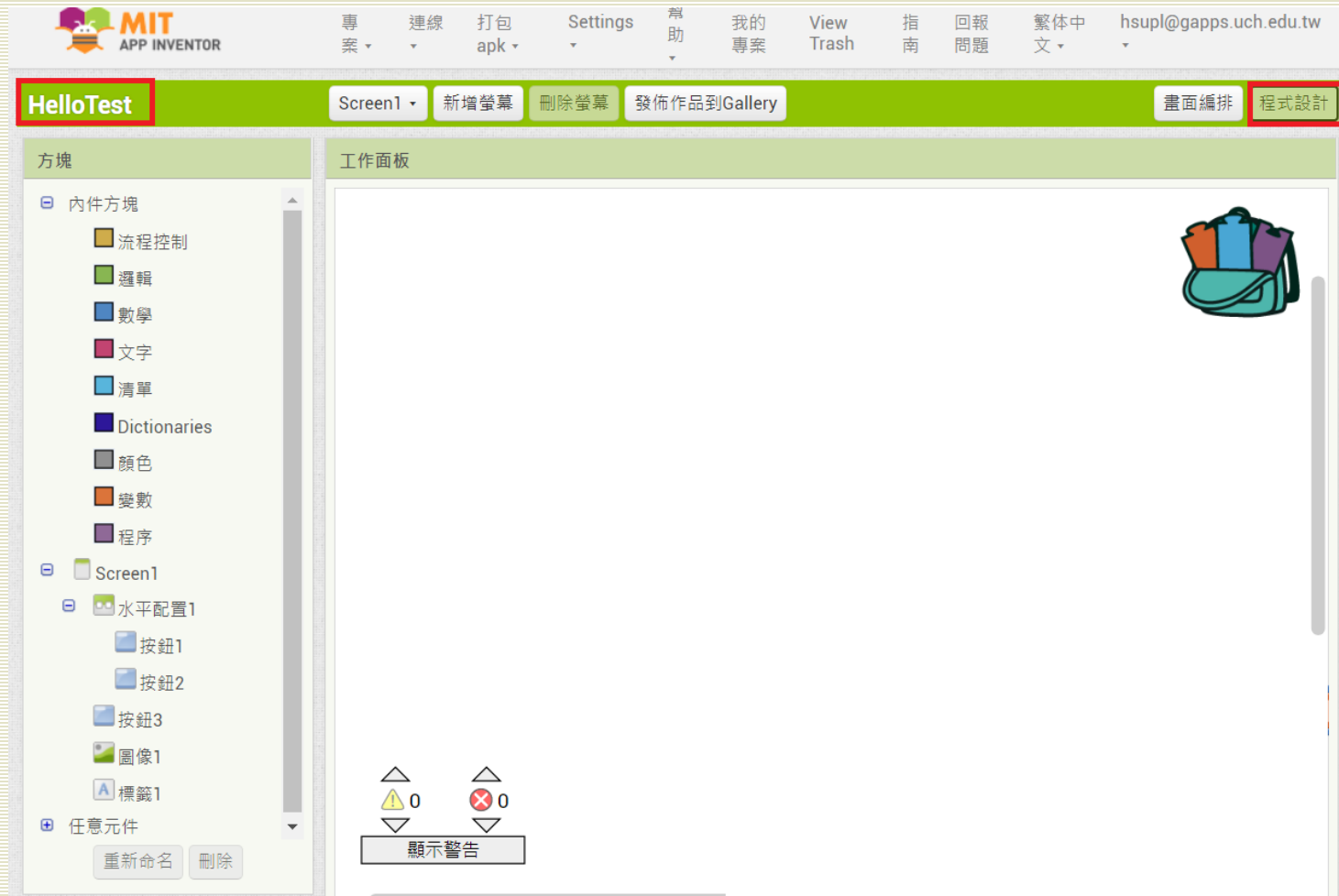
➤ 完成後畫面編排畫面：





Block Editor (堆程式)

➡ 按 程式設計進入 拼圖編輯器 --> 寫程式





Block Editor (堆程式)

◆ 從 Screen1 的 按鈕1 取入:

HelloTest Screen1 新增螢幕 刪除螢幕 發佈作品到Gallery 畫面編排 程式設計

方塊

內件方塊

- 流程控制
- 邏輯
- 數學
- 文字
- 清單
- Dictionaries
- 顏色
- 變數
- 程序

Screen1

- 按鈕1**
- 圖像1
- 標籤1

任意元件

工作面板

當 按鈕1 被點選 執行

當 按鈕1 取得焦點 執行

當 按鈕1 被長按 執行

當 按鈕1 失去焦點 執行

當 按鈕1 被壓下 執行

當 按鈕1 被鬆開 執行

按鈕1 背景顏色

重新命名 刪除





Block Editor (堆程式)

- ◆ 再從 圖像1 取入, 取文字設定 doggie.png

The image displays two screenshots of the Block Editor interface, illustrating the steps to set the image source to 'doggie.png'.

Left Screenshot:

- 方塊 (Blocks):** The '內件方塊' (Inner Blocks) category is expanded, showing '流程控制' (Flow Control), '邏輯' (Logic), '數學' (Math), '文字' (Text), '清單' (List), 'Dictionaries', '顏色' (Color), '變數' (Variable), and '程序' (Script). The '文字' (Text) block is highlighted with a red box.
- 工作面板 (Workspace):** The '當 圖像1 被點選' (When Image 1 is clicked) event block is selected. The '執行' (Do) block contains several actions: '設 圖像1 動畫形式 為' (Set Image 1 animation style to), '圖像1 Clickable', '設 圖像1 Clickable 為' (Set Image 1 Clickable to), '圖像1 高度' (Image 1 height), '設 圖像1 高度 為' (Set Image 1 height to), '設 圖像1 高度百分比 為' (Set Image 1 height percentage to), '圖像1 圖片' (Image 1 image), '設 圖像1 圖片 為' (Set Image 1 image to), and '圖像1 旋轉角度' (Image 1 rotation angle). The '設 圖像1 圖片 為' block is highlighted with a red box.

Right Screenshot:

- 方塊 (Blocks):** The '內件方塊' (Inner Blocks) category is expanded, showing '流程控制' (Flow Control), '邏輯' (Logic), '數學' (Math), '文字' (Text), '清單' (List), 'Dictionaries', '顏色' (Color), and '變數' (Variable). The '文字' (Text) block is highlighted with a red box.
- 工作面板 (Workspace):** The '當 按鈕1 被點選' (When Button 1 is clicked) event block is selected. The '執行' (Do) block contains the action '設 圖像1 圖片 為' (Set Image 1 image to), which is set to 'doggie.png'. A red arrow points from the '文字' block in the '方塊' panel to the 'doggie.png' text in the workspace.

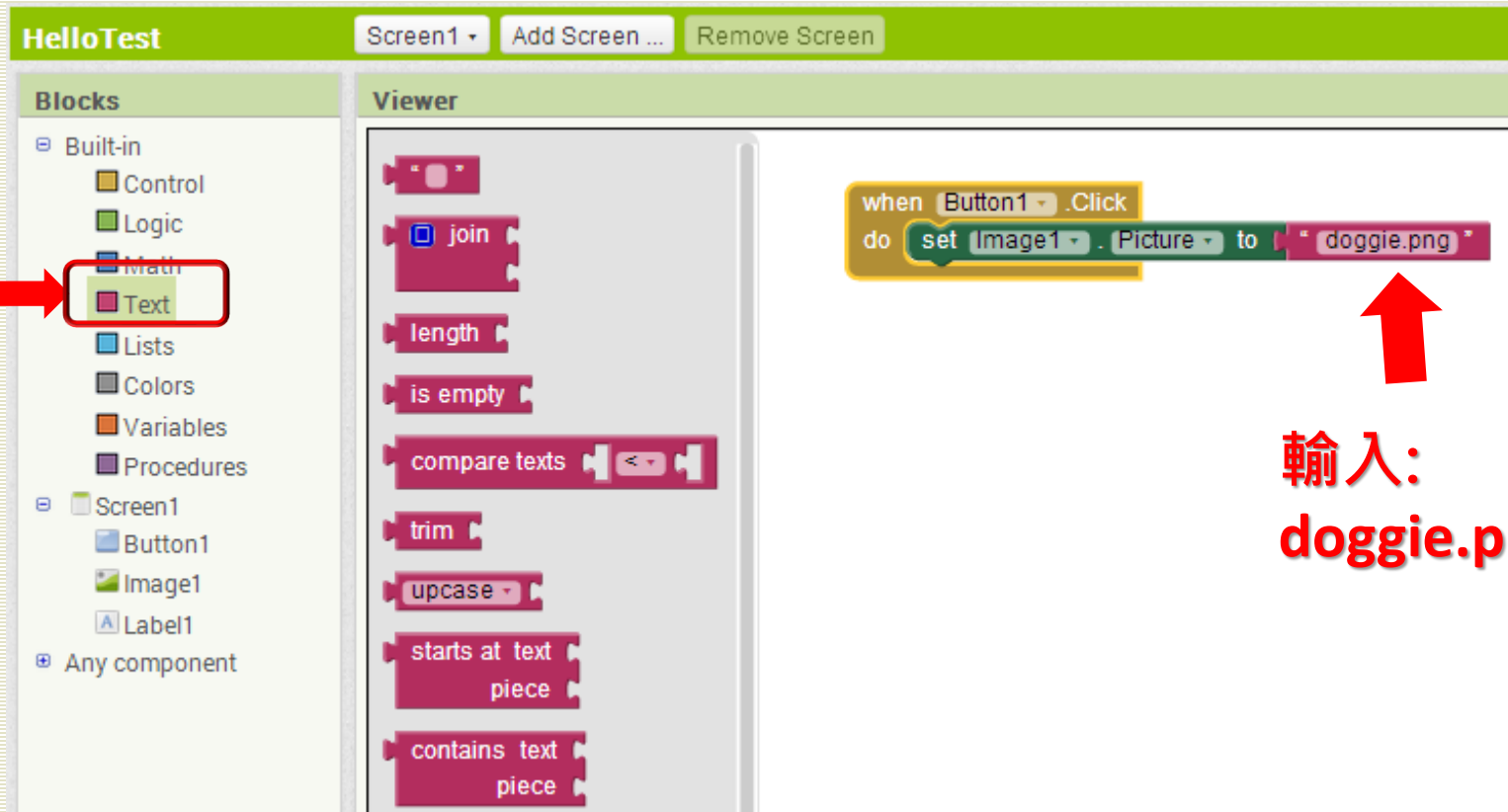




Block Editor (堆程式)

- ◆ 從 Built-In 的 Text 取入: 

按此



The screenshot shows the Block Editor interface. The top bar has 'HelloTest', 'Screen1', 'Add Screen ...', and 'Remove Screen'. The left panel, 'Blocks', lists categories: Built-in (Control, Logic, Math, Text, Lists, Colors, Variables, Procedures), Screen1 (Button1, Image1, Label1), and Any component. The 'Text' block is highlighted with a red box. The right panel, 'Viewer', shows a script: 'when Button1.Click' followed by 'do set Image1.Picture to "doggie.png"'. A red arrow points from the text '輸入: doggie.png' to the 'doggie.png' string in the script.

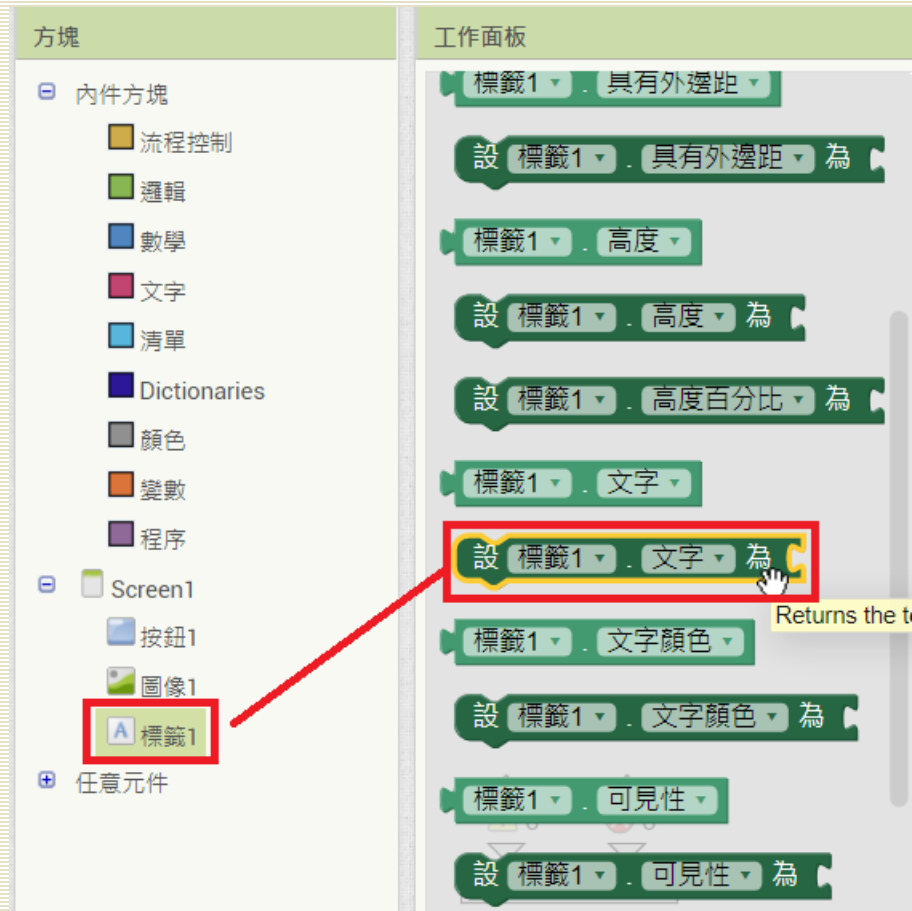
輸入:
doggie.png





Block Editor (堆程式)

◆ 從 Screen1 的 標籤1 取入:





Block Editor (堆程式)

◆ 複製程式方塊

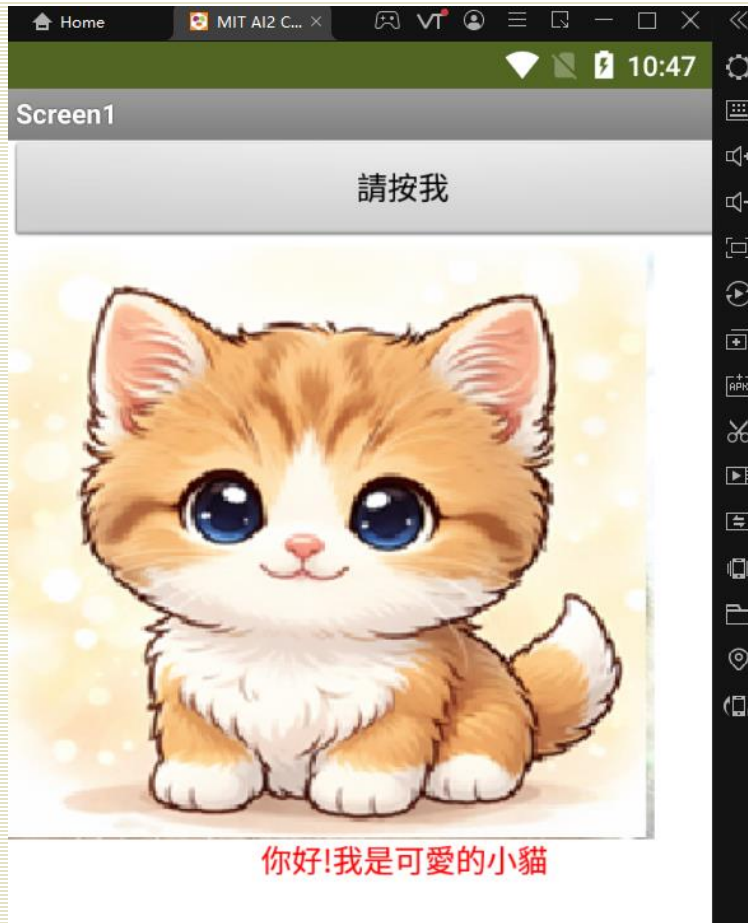
The screenshot illustrates the process of duplicating a code block in the Scratch Block Editor. It shows a sequence of three states:

- Initial State:** A code block with a 'When Button 1 Clicked' trigger, followed by 'Set Image 1 to doggie.png' and 'Set Label 1 to []'.
- Menu State:** A right-click context menu is open over the 'Set Label 1' block, with 'Duplicate Block' selected. Other options include 'Add Comment', 'Collapse Block', 'Disable Block', 'Add to Favourites (0)', 'Delete Block', 'Download Blocks as PNG', 'Help', and 'Run'.
- Intermediate State:** A separate text block containing the text '您好! 我是可愛小狗狗' (Hello! I am a cute little dog) is shown, with a red arrow pointing to it from the text '輸入: 您好! 我是可愛小狗狗' (Input: Hello! I am a cute little dog).
- Final State:** The code block is updated to include both 'Set Image 1 to doggie.png' and 'Set Label 1 to 您好! 我是可愛小狗狗'.





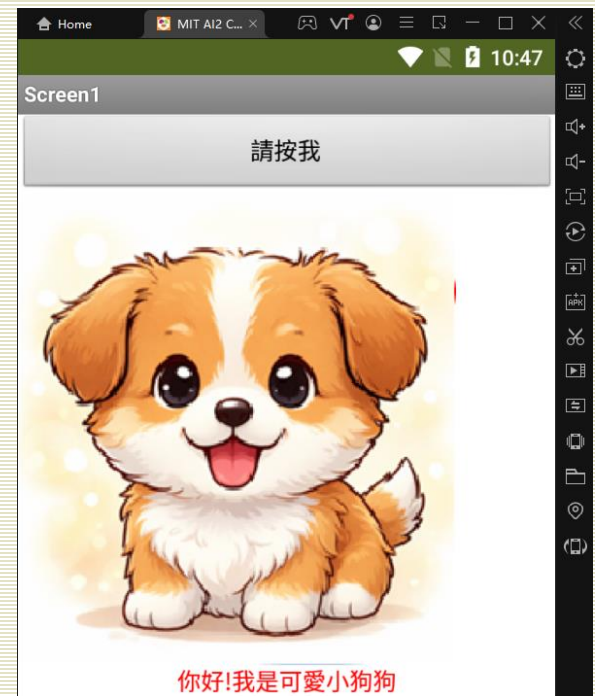
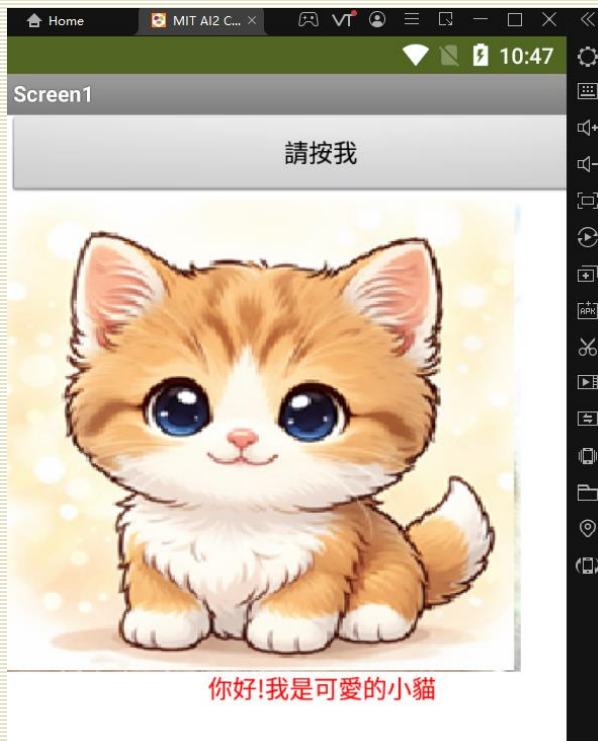
AI Companion (AI 伴侶)





圖片切換功能

- ◆ 請按我可以切換圖片





程式設計

初始化全域變數 toggle 為 1

當 按鈕1 被點選

執行

設置 global toggle 為 取 global toggle + 1

如果 取 global toggle = 1

則 設 Image1 . 圖片 為 "kitty.png"

設 標籤1 . 文字 為 "小貓"

如果 取 global toggle = 2

則 設 Image1 . 圖片 為 "doggie.png"

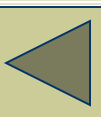
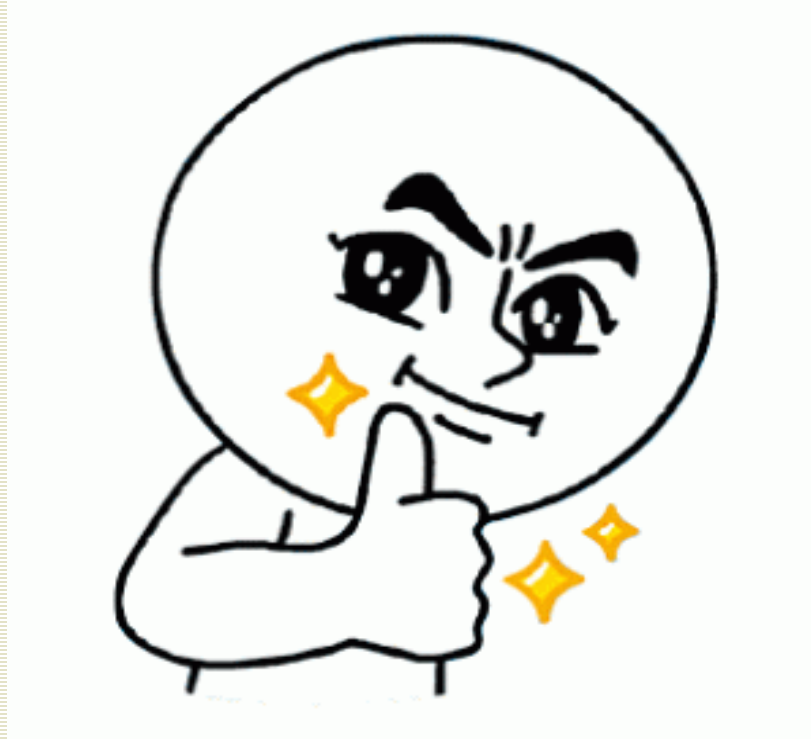
設 標籤1 . 文字 為 "小狗"

設置 global toggle 為 0





實 作 討 論





練習：HelloTest

- 一開始：

貓貓影像出現

貓咪

- 按下請按我：

小狗狗出現了！

小狗





程式設計





作業3：展覽SHOW

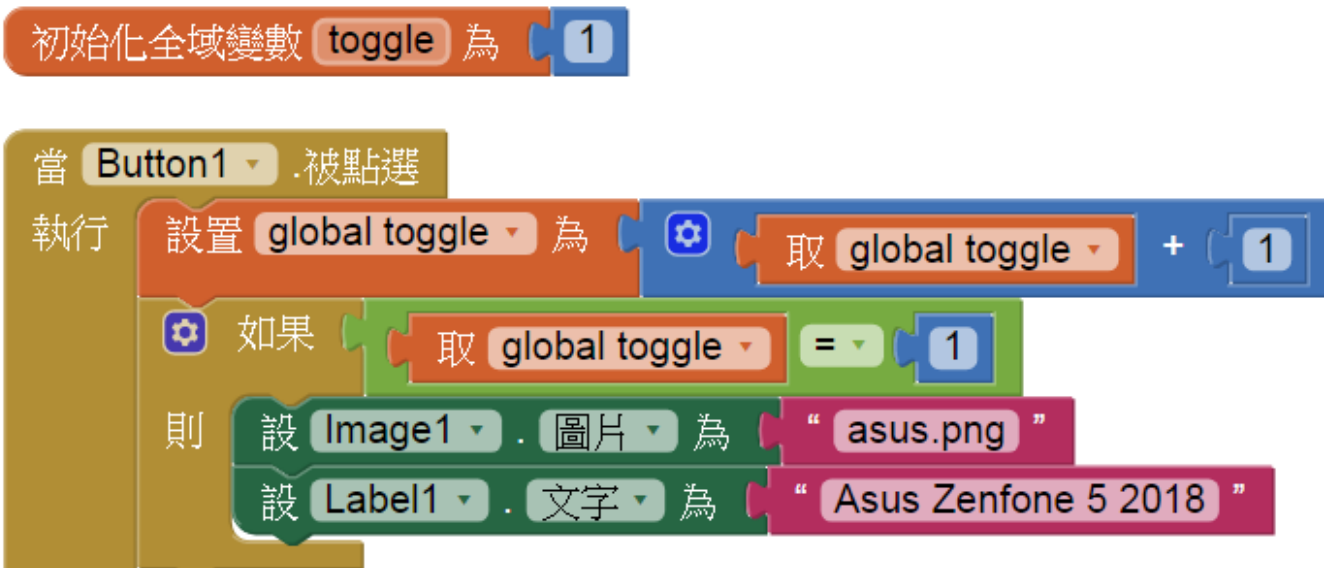
- ◆ 請任選3張照片(同類型)
- ◆ 不可以用手機範例
- ◆ 加上你的學號及姓名





作業3:展覽SHOW

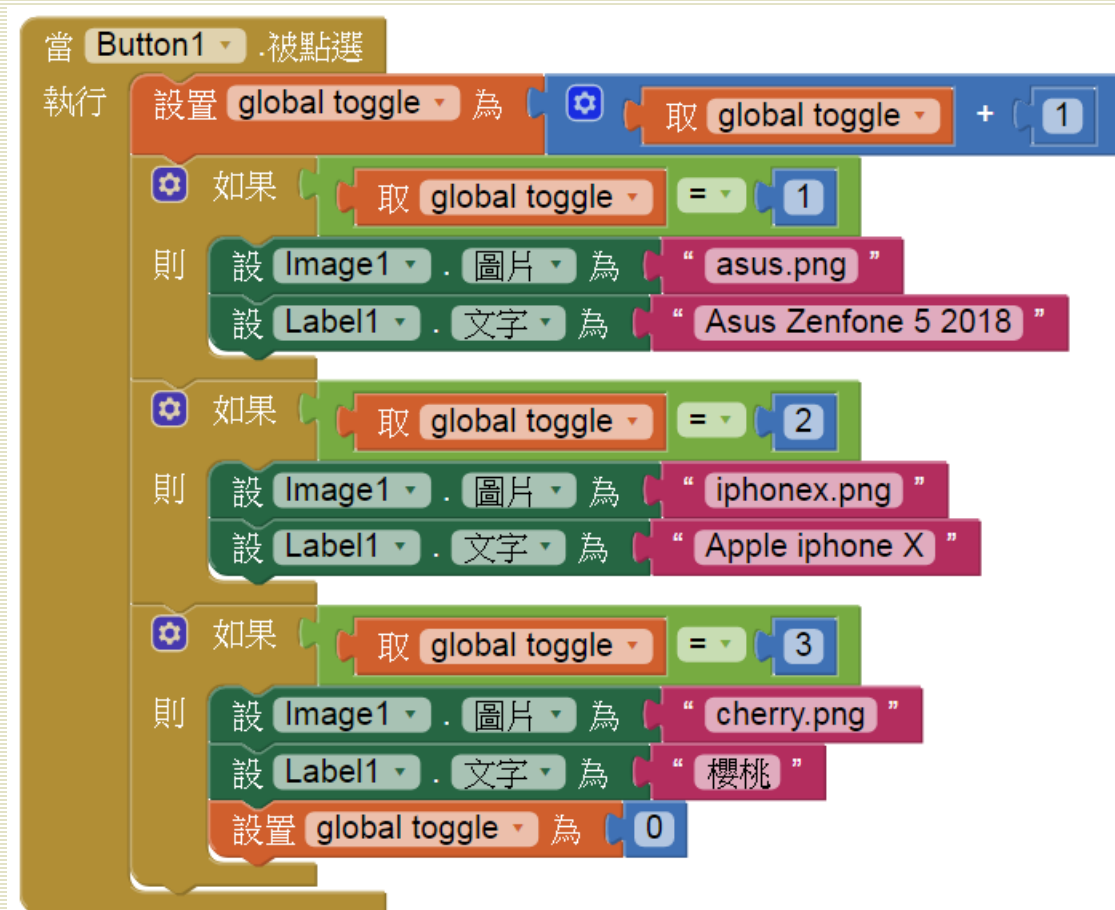
- ◆ 全域變數 toggle (初始化為1, Why? 不是0)
- ◆ 1: asus.png 2: iphonex.png 3. 其他...





作業3:展覽SHOW

- ◆ $\text{toggle} = \text{toggle} + 1$
- ◆ IF $\text{toggle} = 1$ 顯示 asus.png 及文字





作業3：實作結果



Asus Zenfone 5 2018



Apple iPhone X

